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Self-Triggered H_∞ Control Using Projected Actor-Critic-Disturbance Learning

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***Abstract*—Learning-based algorithms for continuous-time differential zero-sum games are promising for cyber–physical systems, yet practical deployment is hindered by two issues: continuous trigger monitoring and possible drift of adaptive weights. This paper develops a self-triggered Actor–Critic–Disturbance learning framework for a class of nonlinear quadratic zero-sum games from a computational-intelligence perspective. Different from periodic or event-triggered schemes, the proposed self-triggered scheduler computes successive update instants using only sampled state information, thus removing continuous state monitoring and reducing communication/computation burden. To enhance learning safety, a projected update law is embedded to confine the adaptive weights of players within predefined compact sets and to prevent instability caused by unbounded learning dynamics. Rigorous analysis establishes Zeno-free execution and guarantees finite-gain $\mathcal{L}_2$ stability for the resulting closed loop. Simulations demonstrate effective regulation, reliable learning behavior, and sparse triggering in the projected H_∞ design.**

***Index Terms*—Self-triggered scheme, Zero-sum game, Projection operator, Actor-Critic-Disturbance learning.**

I. Introduction

Computational intelligence (CI) offers a powerful toolkit for addressing complex decision-making and control problems in nonlinear and potentially adversarial environments [1], [2]. A representative formulation is the zero-sum game (ZSG) that naturally arises in H_∞ control problems among many engineering applications, where two decision-makers have strictly opposing objectives [3], [4]. This makes it essential to design reliable control strategies under adversarial interactions. However, solving a zero-sum game is essentially equivalent to solving the associated Hamilton–Jacobi–Isaacs (HJI) equation, whose analytical solutions are often intractable. Among adaptive control methods [5], [6], [7], [8], [9], [10], [11], adaptive dynamic programming (ADP) [12], [13] has emerged as an effective framework for solving game-theoretic problems [14], [15]. Therefore, in this paper, we solve the zero-sum differential game based on ADP.

I. Related Work

In networked systems, real-time control under limited computational and communication resources is a fundamental challenge [5]. Event-triggered control (ETC) is a resource-efficient alternative to time-triggered strategies. Unlike periodic time-triggered control (TTC), ETC updates only when a state-dependent condition is met, reducing unnecessary computation and network traffic [6], [16]. This aligns with CI principles such as real-time adaptability and robustness to uncertainty [6], [17]. Recent progress further promotes the fusion of ETC with data-driven learning and robustness enhancement: data-driven event-triggered ADP (ETADP) integrates learning and triggering to optimize control performance and resource usage under practical constraints (e.g., input saturation) [18], while extended disturbance-observer-based data-driven designs leverage event-triggered outputs to estimate and reject unknown disturbances in networked nonlinear systems [19]. These advances reinforce ETC as a CI-enabled mechanism for resource-efficient and adaptive control. In the context of ZSG, ETADP has enabled learning-based triggering mechanisms that improve both control performance and resource usage [17], [20], [21], [22]. Based on the state evolution of the system, ETC strategy makes it feasible to apply ADP in resource-constrained environments by determining when to update control input in an adversarial setting [11], [17], [23]. A major limitation of ETC is its reliance on continuous state measurement to evaluate triggering conditions [24]. This persistent monitoring can incur substantial computational and communication overhead, undermining practicality under bandwidth constraints [24], [25], [26]. Hence, alternatives are needed that preserve the benefits of ETC while mitigating these inherent costs.

In contrast to ETC methods, self-triggered control (STC) schemes emerge as a more efficient alternative [25], [26]. The main idea behind STC is to solve a transcendental equation to determine future triggering moments [27], guaranteeing that system states are updated only when necessary [26]. This eliminates the reliance on continuous state monitoring, significantly reducing computational and communication costs [24]. The self-triggered ADP (STADP) therefore offers a resource-efficient and intelligent control framework [24]. STC

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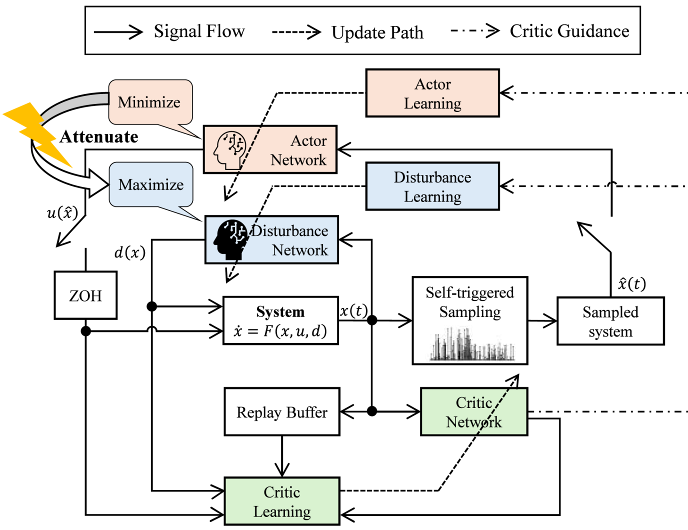

Fig. 1. The block diagram of the self-triggered ACD learning approach, where $x(t)$ is the system state, $\hat{x}(t)$ is the sampled state, u is the control input (minimizing player), d is the disturbance input (maximizing player).

guarantees efficient update while ADP provides an adaptive online learning structure with real-time disturbance attenuation within the ZSG formulation. This enhances control performance and feasibility in resource-constrained settings [24], [25]. In relation to online learning, properties such as stability, convergence, and boundedness of adaptive weights are factors that must be addressed in designing optimization strategies. With regard to existing ADP-based control methods, such as adaptive Actor-Critic learning [24], [28], policy iteration [12], [29], and Barrier-Critic learning [30], [31], but the discussions on adaptive weights subject to constraints remains insufficient.

To effectively guarantee the boundedness of learning parameters in ADP, the projection operator (PO) has been introduced as a powerful practical tool [32], [33]. Since ADP learning laws are typically gradient-driven and operate online under imperfect excitation, approximation errors, disturbances, etc., any of which can cause parameter drift in adaptive weights, this highlights the necessity of the PO [20], [24]. The basic principle of projected learning is to explicitly constrain adaptive weights to a predefined compact set, so that parameter trajectories remain bounded even in the presence of worst-case disturbance. In each update step, the PO acts only when the update would move the weights outside the admissible region. It then projects the update back onto the boundary to enforce the constraints without otherwise altering the learning law [20], [24], [32], [33], [34]. Therefore, by keeping the adaptive weights within a safe region, the PO provides a safeguard that makes the ZSG-oriented ADP implementation more reliable and helps preserve the closed-loop stability.

Contribution: Learning-based H_∞ designs for resource-limited nonlinear ZSGs remain an open challenge in the current landscape, where most H_∞ ETADP schemes still require continuous monitoring and are vulnerable to online parameter drift that can compromise safety. This challenge is addressed in this paper via a self-triggered and safety-aware ADP framework. As depicted in Fig. 1, the projected Actor-Critic-Disturbance (ACD) network provides the learning structure with the state sampling process and a zero-order hold (ZOH). This paper advances the state of the art in ADP for ZSGs by introducing a projected STADP framework that eliminates continuous monitoring while preserving H_∞ robustness, with the following main contributions.

1) Compared with prior ETADP designs [20], [22] that lead to continuous monitoring, the problem of determining execution instants is transformed into solving a transcendental equation in this paper. We prove that a unique solution to the transcendental equation exists, which also ensures convergence to the Nash equilibrium.
2) Existing learning-based ADP methods [17], [35] do not explicitly constrain the adaptive weights, so parameter drift under weak excitation or inappropriate tuning parameters may destabilize learning dynamics. To tackle this safety concern, we incorporate POs into the learning laws to prevent unbounded weight evolution without increasing computational complexity.

Structure: The remainder of the paper is organized as follows. In Section II, mathematical preliminaries are provided. Section III describes the ZSG and H_∞ control problem. Discussions on the ETC and STC are conducted in Section IV. The self-triggered ACD learning is proposed in Section V, followed by simulation verification in Section VI. Conclusions follow in Section VII.

Notations and Terminologies: Throughout this paper, "iff" and "s.t." denote "if and only if" and "such that," respectively. "$\triangleq$" denotes equality by definition. Let $\mathbb{N}$ and $\mathbb{R}$ be the sets of natural and real numbers, with $\mathbb{N}_0 \triangleq \mathbb{N} \cup 0$, $\mathbb{R}^{n\times m}$ the set of $n \times m$ real matrices and $\mathbb{R}^n \triangleq \mathbb{R}^{n\times 1}$ the n-dimensional Euclidean space. For $A \in \mathbb{R}^{n\times m}$, $A^\top$ denotes its transpose. The weighted Euclidean norm is $\|x\|_R \triangleq (x^\top R x)^{1/2}$, and $\|x\|$ denotes the unweighted case ($R = I$). $\bar{\lambda}(A)$ and $\underline{\lambda}(A)$ denote the maximum and minimum eigenvalues of a square matrix A, respectively. The indicator $\mathbf{1}\{\phi\}$ equals 1 if ϕ is true and 0 otherwise. A set $\Omega \subset \mathbb{R}^n$ is compact if it is closed and bounded, with interior Ω° and boundary $\partial\Omega$. A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is C^1 if it is continuously differentiable, with gradient ∇f and transposed gradient $\nabla^\top f$. A signal $y : [0, \infty) \to \mathbb{R}^n$ is $\mathcal{L}_2$ if $\|y\|_{\mathcal{L}_2} \triangleq \sqrt{\int_0^\infty \|y(t)\|^2 \,\mathrm{d}\,t}$ is finite.

II. Preliminaries

A. Finite-Gain $\mathcal{L}_2$ Stability

Consider the nonlinear dynamical system

$$\begin{aligned} \dot{x} &= f_\mathrm{c}(x) + k(x)d, \quad x(0) = x_0, \\ y &= h_\mathrm{c}(x), \end{aligned} \tag{1}$$

where $x(t) \in \mathbb{R}^n$ is the state, $d(t) \in \mathbb{R}^{n_d}$ the disturbance input, and $y(t) \in \mathbb{R}^{n_y}$ the output of the system, at time $t \geq 0$; $x_0 \in \mathbb{R}^n$ is an initial state; locally Lipschitz continuous functions $f_\mathrm{c}(x) \in \mathbb{R}^n$ and $k(x) \in \mathbb{R}^{n\times n_d}$ represent the (closed-loop) drift and disturbance dynamics, respectively; $h_\mathrm{c}(x)$ is a continuous (closed-loop) output function. We suppose that (i) f_c and h_c vanish at the origin, i.e., $f_\mathrm{c}(0) = 0$ and $h_\mathrm{c}(0) = 0$, and that (ii) there exist positive constants c_x, c_k and c_h s.t.

$$\|k(x)\| \leq c_k \text{ and } \|h_\mathrm{c}(x)\| \leq c_h\|x\| \quad \text{whenever } \|x\| \leq c_x.$$

Definition 1 (Finite-gain $\mathcal{L}_2$ stable [36]): The system (1) is said to be finite-gain $\mathcal{L}_2$ stable iff there exist constants $\gamma \geq 0$ and $b \geq 0$ s.t.

$$\|y\|_{\mathcal{L}_2} \leq \gamma \|d\|_{\mathcal{L}_2} + b, \qquad \forall d \in \mathcal{L}_2. \tag{2}$$

Definition 2 (Available Storage, [36]): For the system (1) with supply rate $s(d,y)$, the available storage V_a is defined as

$$V_\mathrm{a}(x_0) \triangleq \sup_{d\in\mathcal{L}_2, T\geq 0} \int_0^T -s(d(t),y(t))\,\mathrm{d}\,t. \tag{3}$$

The quantity $V_\mathrm{a}(x_0)$ can be interpreted as the maximum energy that can be extracted from the system (1). Additionally, the available storage can be employed to analyze Lyapunov stability with zero-state observability (ZSO) [36], as discussed in [36, Lemma 3.2.3]. The connection between Lyapunov stability of the unforced system and finite-gain $\mathcal{L}_2$ stability of the input-output system is referred to [37, Theorem 5.1].

B. Projected Gradient Descent

Definition 3 (Type-I Projection [33]): Given $\theta, y \in \mathbb{R}^n$, let

$$\mathrm{Proj}_\mathrm{I}(y, \theta \,|\, \mathcal{G}, \Theta, \Gamma) \triangleq y - \mathbf{1}\{\theta \in \Theta\} \cdot \Gamma \frac{\nabla \mathcal{G}(\theta)\, \nabla^\top \mathcal{G}(\theta)}{\|\nabla \mathcal{G}(\theta)\|_\Gamma^2} y$$

for $\Gamma \in \mathbb{R}^{n\times n}$, $\Theta \subset \mathbb{R}^n$, and C^1 convex function $\mathcal{G} : \mathbb{R}^n \to \mathbb{R}$.

With the Type-I projection, constraint satisfaction can be enforced using projected gradient descent, as discussed below.

Lemma 1: Consider the minimization problem

$$\min\nolimits_\theta J(\theta) \;\text{ subject to }\; \theta \in \Psi \triangleq \{\theta \in \mathbb{R}^n \,|\, \mathcal{G}(\theta) \leq 0\}$$

w.r.t. given convex functions $J, \mathcal{G} \in \mathrm{C}^1$, with its optimal solution $\theta^* \triangleq \arg\min_{\theta\in\Psi} J(\theta) \in \Psi$. Then, under the continuous projected gradient descent adaptation law:

$$\dot{\theta} = \mathrm{Proj}_\mathrm{I}\left(y, \theta \mid \mathcal{G}, \Omega_\mathcal{G}, \Gamma\right), \quad \theta(0) = \theta_0 \in \Psi, \tag{4}$$

with $\Gamma \succ 0$ being sufficiently small, $y \triangleq -\Gamma \nabla J(\theta)$, and

$$\Omega_\mathcal{G} \triangleq \left\{\theta \in \mathbb{R}^n \mid \mathcal{G}(\theta) = 0 \text{ and } y^\top \nabla \mathcal{G}(\theta) > 0\right\}$$

(i) $\theta(t)$ remains within Ψ for all $t \geq 0$; (ii) $\lim\limits_{t\to\infty} \theta(t) = \theta^*$.

Proof: First, consider the time derivative $\dot{\mathcal{G}}(\theta) = \dot{\theta}^\top \nabla \mathcal{G}(\theta)$. If $\theta \in \Psi^\mathrm{o}$, then $\theta \in \Psi$ trivially holds. In addition, under the projected adaptation law (4) and for $\theta \in \partial\Psi$ (i.e., $\mathcal{G}(\theta) = 0$) and $y \triangleq -\Gamma \nabla J(\theta)$,

- $y^\top \nabla \mathcal{G}(\theta) \leq 0 \Longrightarrow \dot{\mathcal{G}}(\theta) = y^\top \nabla \mathcal{G}(\theta) \leq 0$ trivially;
- $y^\top \nabla \mathcal{G}(\theta) > 0 \Longrightarrow \dot{\mathcal{G}}(\theta) = 0$, thanks to the projection.

Therefore, $\dot{\mathcal{G}}(\theta) \leq 0$ and thus $\mathcal{G}(\theta(t))$ does not increase over time. That is, $\theta(t)$ remains in Ψ for all $t \geq 0$ if $\theta_0 \in \Psi$.

Next, define $\tilde{\theta} \triangleq \theta - \theta^*$; consider the Lyapunov candidate $V(\tilde{\theta}) = \frac{1}{2}\tilde{\theta}^\top \Gamma^{-1} \tilde{\theta}$, of which the time derivative is $\dot{V}(\tilde{\theta}) = \tilde{\theta}^\top \Gamma^{-1} \mathrm{Proj}_\mathrm{I}(y, \theta \mid \mathcal{G}, \Omega_\mathcal{G}, \Gamma)$. Then, the asymptotic convergence of θ to θ^* is established by Lyapunov's Theorem, with $\dot{V}$ being negative definite as follows.

(Case I): If $\theta \notin \Omega_\mathcal{G}$, then $\dot{V}(\tilde{\theta}) = -\tilde{\theta}^\top \nabla J(\theta)$. By the first-order convexity condition,

$$\dot{V}(\tilde{\theta}) = -(\theta - \theta^*)^\top \nabla J(\theta) \leq J(\theta^*) - J(\theta) \leq 0$$

and thus $\dot{V}(\tilde{\theta}) = 0$ only if $\theta = \theta^*$, i.e., $\tilde{\theta} = 0$. Therefore, we conclude in this case that $\dot{V}$ is negative definite.

(Case II): If $\theta \in \Omega_\mathcal{G}$, we have $\dot{V}(\tilde{\theta}) = -\tilde{\theta}^\top \nabla J(\theta) - W_\theta$,

$$\text{where } W_\theta \triangleq \frac{\tilde{\theta}^\top \nabla \mathcal{G}(\theta)}{\|\nabla \mathcal{G}(\theta)\|_\Gamma^2} \cdot \nabla^\top \mathcal{G}(\theta) y.$$

Since $\tilde{\theta}^\top \nabla \mathcal{G}(\theta) \geq 0$ for $\theta^* \in \Psi$, thanks to the convexity of $\mathcal{G}$, $W_\theta \geq 0$ holds whenever $\nabla^\top \mathcal{G}(\theta) y > 0$. I.e., W_θ can make $\dot{V}(\tilde{\theta})$ only more negative, and thus $\dot{V}$ is negative definite. □

The operator $\mathrm{Proj}_\mathrm{I}(\cdot)$ forces the parameter trajectory $\theta(t)$ to remain in Ψ even when the gradient step would violate the constraint, which prevents unbounded updates.

Definition 4 (Type-II Projection [32]): For $\theta \in \mathbb{R}^n$ and $\rho > 0$,

$$\mathrm{Proj}_\mathrm{II}(\theta \,|\, \rho) \triangleq \theta + \mathbf{1}\{\rho < \|\theta\|\} \cdot \frac{\rho - \|\theta\|}{\|\theta\|} \theta.$$

The discrete-time projected gradient descent based on this type-II projection, and its boundedness, are given as follows.

Lemma 2: [24], [32] Consider the minimization problem

$$\min_\theta E(\theta) \;\text{ subject to }\; \theta \in \Phi \triangleq \{\theta \in \mathbb{R}^n \,|\, \|\theta\| \leq \rho\}$$

w.r.t. given convex objective function $E \in \mathrm{C}^1$ and $\rho > 0$, with its ideal solution $\theta^* \triangleq \arg\min_{\theta\in\Phi} E(\theta) \in \Phi$. Then, with sufficiently small $\alpha > 0$, under projected gradient descent law:

$$\theta_{k+1} = \mathrm{Proj}_\mathrm{II}(\hat{\theta}_k \,|\, \rho), \; \hat{\theta}_k = \theta_k - \alpha \nabla E(\theta_k), \; \theta_0 \in \Phi,$$

(i) θ_k remains in Φ for all $k \in \mathbb{N}$; (ii) $\lim_{k\to\infty} \theta_k = \theta^*$; (iii) $(\bar{\theta}_k - \theta^*)^\top (\hat{\theta}_k - \bar{\theta}_k) \geq 0$ for all $k \in \mathbb{N}$ and $\bar{\theta}_k \triangleq \rho \cdot \hat{\theta}_k / \|\hat{\theta}_k\|$.

The operator $\mathrm{Proj}_\mathrm{II}(\cdot)$ enforces the norm bound $\|\theta_k\| \leq \rho$ at every iteration, so θ_k stays in Φ for all k. The associated inequality $(\bar{\theta}_k - \theta^*)^\top (\hat{\theta}_k - \bar{\theta}_k) \geq 0$ implies that projection does not increase the distance to the optimizer and supports convergence [32], [33].

III. Problem Statement

Consider the continuous-time nonlinear system

$$\dot{x} = \underbrace{f(x) + g(x)u + k(x)d}_{\triangleq F(x,u,d)}, \quad x(0) = x_0,$$
$$y = h(x,u), \tag{5}$$

where $x(t) \triangleq [x_1(t), \ldots, x_n(t)]^\top$ is the state, $u(t) \in \mathbb{R}^{n_u}$ the control input and $d(t) \in \mathbb{R}^{n_d}$ the disturbance input with finite energy. The functions $f \in \mathbb{R}^n$, $g \in \mathbb{R}^{n\times n_u}$ and $k \in \mathbb{R}^{n\times n_d}$ representing the drift, input and disturbance dynamics, respectively, are known and locally Lipschitz continuous with $f(0) = 0$ [14]. $y(t) \in \mathbb{R}$ is the fictitious performance output determined at time t by the output function h given by

$$\|h(x,u)\|^2 = Q(x) + \|u\|_R^2 \tag{6}$$

for continuous function $Q(x) \succ 0 \in \mathbb{R}$ and symmetric matrix $R \succ 0 \in \mathbb{R}^{n_u\times n_u}$. Moreover, for well-definedness, $Q(x)$ is radially unbounded on a compact set $\Omega \subset \mathbb{R}^n$(i.e., $\exists\, q_2 \geq q_1 > 0$ s.t. $q_1\|x\|^2 \leq Q(x) \leq q_2\|x\|^2, \forall x \in \Omega$) and $\bar{\lambda}(R) < \infty$ [28].

By Lipschitz continuity and $f(0) = 0$, $\exists\, b_f > 0$ s.t.

$$\|f(x)\| \leq b_f\|x\|, \quad \forall x \in \Omega. \tag{7}$$

Moreover, since Ω is compact, g and k are bounded on Ω, i.e., there exist $b_g, b_k > 0$ s.t.

$$\|g(x)\| \leq b_g \text{ and } \|k(x)\| \leq b_k, \quad x \in \Omega. \tag{8}$$

We suppose that the system (5) is stabilizable over the compact set Ω when $d(t) \equiv 0$ and satisfies the ZSO condition [36]:

$$u(t) \equiv 0, d(t) \equiv 0, \text{ and } y(t) \equiv 0 \text{ implies } x(t) \equiv 0. \tag{9}$$

Define the performance index [14] associated with (5) as

$$J(u, d; \gamma, x_0) = \int_0^\infty \ell(x(t), u(t), d(t))\mathrm{d}t$$

with the instantaneous cost $\ell(x, u, d) \triangleq \|h(x, u)\|^2 - \gamma^2\|d\|^2$.

Problem 1 (Disturbance Attenuation): For system (5), the disturbance attenuation problem aims to design the control input $u \in \mathcal{U}$ such that

$$\sup_{d \in \mathcal{L}_2} \frac{\|h(x, u)\|_{\mathcal{L}_2}}{\|d\|_{\mathcal{L}_2}} \leq \gamma \tag{10}$$

given attenuation level $\gamma \geq \gamma^* > 0$, where γ^* is the smallest attenuation level s.t. the system is stabilizable; $\mathcal{U}$ denotes the set of all locally Lipschitz policies $u(x) \in \mathbb{R}^{n_u}$ s.t. $u(0) = 0$ and the system (5) is (i) asymptotically stable when $d(t) \equiv 0$; (ii) finite-gain $\mathcal{L}_2$ stable w.r.t. any disturbance input $d \in \mathcal{L}_2$ and the corresponding performance output $y = h(x, u)$.

For Problem 1, for all feedback inputs $u \in \mathcal{U}$ and $d \in \mathcal{L}_2$, define the value function of $\{u, d\}$ as

$$V(x; u, d) = \int_t^\infty \ell(x(\tau), u(x(\tau)), d(x(\tau)))\mathrm{d}\tau, \tag{11}$$

which solves the Lyapunov-like equation:

$$\mathcal{H}(x, u, d, \nabla V) = 0 \tag{12}$$

where the Hamiltonian $\mathcal{H}$ is defined as

$$\mathcal{H}(x, u, d, \nabla V) \triangleq \ell(x, u, d) + \nabla^\top V(x) F(x, u, d). \tag{13}$$

According to the dynamic game theory [3], [4], [38], the disturbance attenuation problem (10) can be converted into the zero-sum differential game

$$\min_{u \in \mathcal{U}} \max_{d \in \mathcal{L}_2} J(u, d; \gamma, x_0) \tag{14}$$

with the minimizing player u and the maximizing player d.

The *outcome* [3] of the ZSG (14) is given by

$$V^*(x) = \min_{u \in \mathcal{U}} \max_{d \in \mathcal{L}_2} J(u, d; \gamma, x_0).$$

Moreover, the ZSG (14) admits a unique solution $\{u^*, d^*\}$, if the following Nash condition [3], [4], [14] holds:

$$\min_{u \in \mathcal{U}} \max_{d \in \mathcal{L}_2} J(u, d; \gamma, x_0) = \max_{d \in \mathcal{L}_2} \min_{u \in \mathcal{U}} J(u, d; \gamma, x_0). \tag{15}$$

Enforcing the stationary conditions [14]:

$$\frac{\partial H(x, u, d, \nabla V^*)}{\partial u} = 0 \text{ and } \frac{\partial H(x, u, d, \nabla V^*)}{\partial d} = 0$$

yields the Nash equilibrium:

$$u^*(x) = -\frac{1}{2} R^{-1} g^\top(x) \nabla V^*(x),$$

$$d^*(x) = \frac{1}{2\gamma^2} k^\top(x) \nabla V^*(x). \tag{16}$$

It should be noted that the best maximizing player d^* characterizes the worst-case disturbance at the Nash equilibrium of the ZSG (14). The best minimizing player u^* aims to attenuate d^* in the H_∞ sense and thus attenuates any disturbance in $\mathcal{L}_2$.

Substituting the Nash equilibrium $\{u^*, d^*\}$ in (16) into (12) gives the HJIE [4], [38]:

$$\begin{aligned} 0 &= \mathcal{H}(\cdot, u^*, d^*, \nabla V^*) \\ &= Q + \nabla^\top V^* f - \frac{1}{4} \nabla^\top V^* g R^{-1} g^\top \nabla V^* \\ &\quad + \frac{1}{4\gamma^2} \nabla^\top V^* k k^\top \nabla V^* \end{aligned} \tag{17}$$

for all $x \in \Omega$ with $V^*(0) = 0$.

In what follows, suppose the HJIE (17) admits a unique positive semi-definite solution $V^* \in \mathrm{C}^1$. Then, by [36, Lemma 3.2.3] and the ZSO condition (9): (i) V^* is positive definite; (ii) the equilibrium $x = 0$ of the system (5) under u^* is asymptotically stable when $d \equiv 0$; (iii) the asymptotic stability is global under:

Assumption 1: [37] V^* is radially unbounded, s.t. $c_1\|x\|^2 \leq V^*(x) \leq c_2\|x\|^2$ for constants $c_2 \geq c_1 > 0$.

For well-definedness and technical reasons, we further assume Lipschitz continuity of $u \in \mathcal{U}$, so that $\exists\, \mathcal{L}_u > 0$ s.t.

$$\|u^*(x) - u^*(x')\| \leq \mathcal{L}_u \|x - x'\|, \qquad \forall x, x' \in \Omega. \tag{18}$$

The H_∞ control u^* and the worst-case disturbance d^* depend on the continuous state $x(t)$ that needs to be constantly measured and updated [15].

IV. H_∞ Control With Intermittent Feedback

A main concern of this paper is to develop a sampled-data implementation of the minimizing player u^* in (16) for the zero-sum differential game (14), which ensures the finite-gain $\mathcal{L}_2$ stability for the system (5) in the presence of the worst-case disturbance player d^* in (16).

First, the sampled-state model is introduced in Section IV-A. Second, the ETC baseline is briefly revisited in Section IV-B. Third, in Section IV-C, the novel self-triggered sampling rule is developed with feasibility and stability analysis.

A. Sampled-State Model

To reduce computational overhead and communication load, we employ a synchronous intermittent feedback strategy based on the aperiodically sampled full-state vector $\hat{x}(t)$ (i.e., all sub-states being sampled at the same time t [25], [26], [35]). Specifically, all state components are sampled simultaneously and held constant over execution intervals, i.e.,

$$\hat{x}(t) = x(t_i) \text{ for } t \in [t_i, t_{i+1}) \tag{19}$$

for an increasing sequence of sample instants $t_i \geq 0$ that starts from $t_0 = 0$. $\tilde{x} \triangleq x - \hat{x}$ denotes the associated sample error.

The synchronous construction (19) ensures that the control input using the feedback $\hat{x}(t) = [\hat{x}_1(t), \ldots, \hat{x}_n(t)]^\top$ can be

TABLE I
COMPARISON OF EXISTING ADP

Ref	Scheme	Monitor	PO	f	g	k
[20]	ETC	Yes	Yes	U	K	-
[22]	ETC	Yes	No	U	K	K
[24]	STC	No	Yes	K	K	-
[25]	STC	No	No	K	K	-
[39]	No	-	Yes	U	K	-
[40]	No	-	No	U	K	-
Ours	STC	No	Yes	K	K	K

Note: K = known; U = unknown; Monitor = continuous monitoring of the triggering condition.

well-defined and implementable through a ZOH, s.t. the controller maps $\hat{x}$ onto a continuous-time input signal:

$$\hat{u}^*(t) \triangleq u^*(\hat{x}) = -\frac{1}{2}R^{-1}g^\top(\hat{x})\nabla V^*(\hat{x}). \tag{20}$$

The algebraic property of the Hamiltonian $\mathcal{H}$ with intermittent sampled state $\hat{x}$ in $\hat{u}^*$ differs from that under continuous feedback as summarized below.

Lemma 3: [22] The sample-based Hamiltonian satisfies

$$\mathcal{H}(x, u^*(\hat{x}), d^*(x), \nabla V^*(x)) = \|u^*(x) - u^*(\hat{x})\|_R^2.$$

The following standard assumption on the system dynamics F in (5) is required for subsequent discussions.

Assumption 2: [25] $\forall\, u \in \mathcal{U}$, $d \in \mathcal{L}_2$ and $\forall\, x \in \Omega$ with the sampling error $\tilde{x}$ under (19), $\exists\, K > 0$ s.t.

$$x^\top F(x, u(\hat{x}), d(x)) \le K(1 + \|x\|^2 + \|\tilde{x}\|^2), \tag{21}$$

$$\tilde{x}^\top F(x, u(\hat{x}), d(x)) \le K(1 + \|x\|^2 + \|\tilde{x}\|^2). \tag{22}$$

Remark 1: Assumption 2 implies a quadratic growth bound on the system (5) under sampled-state feedback, which guarantees forward completeness and thus avoids finite escape. Moreover, it yields computable bounds for both the state and the sampling error, which are instrumental for constructing the self-triggered sampling rule later in Section IV-C.

As summarized in Table I, existing results on intermittent feedback schemes can be categorized into two types, namely ETC and STC. The paper designs a STADP scheme that determines the next update instant in advance, thereby avoiding continuous state monitoring, which overcomes the shortcomings in ETADP as discussed in the following subsections.

B. Event-Triggered Framework

Based on the aforementioned sampled-state model (19), we now briefly introduce the conventional ETC design with its properties detailed in the following lemma.

Lemma 4: [22] Consider the sampling-based control input $\hat{u}^*$ in (20) and the external disturbance d^* in (16) under the triggering condition for $q_1 > 0$ and $\alpha \in (0, 1)$:

$$\|\tilde{x}\|^2 \le \frac{q_1\alpha\|x\|^2 + \|\hat{u}^*\|_R^2 - \gamma^2\|d^*\|^2}{2\mathcal{L}_u^2\bar{\lambda}(R)}. \tag{23}$$

Then, the value function $V(\,\cdot\,; \hat{u}^*, d^*)$ under the event-triggered mechanism can be expressed as

$$V(x_0; \hat{u}^*, d^*) = \int_0^\infty \|u^*(t) - \hat{u}^*(t)\|_R^2 \,\mathrm{d}\,t + V(x_0; u^*, d^*)$$

with the closed-loop system being asymptotically stable.

The lemma indicates that ETC achieves asymptotic stability, at the expense of degraded performance (i.e., increased cost). However, enforcing (23) requires continuously monitoring the time-varying signals $\tilde{x}(t)$, $x(t)$, and $d^*(x(t))$, which can impose substantial communication overhead. This motivates a more communication-efficient sampling rule as discussed next.

C. Self-Triggered Framework

The following lemma relates the upper and lower bounds of the norms of $\tilde{x}(t)$ and $x(t)$ to the norm of the last-observed state $\hat{x}$, respectively.

Lemma 5: [24], [25] Let Assumption 2 hold, and the system state be updated as (19). Then, $\forall t \in [t_i, t_{i+1})$ and $\forall i \in \mathbb{N}_0$,

$$\|\tilde{x}(t)\|^2 \le \Im_1(\|\hat{x}\|, t - t_i), \tag{24}$$

$$\|x(t)\|^2 \ge \Im_2(\|\hat{x}\|, t - t_i), \tag{25}$$

where $\Im_1(r, t) \triangleq \frac{2r^2+1}{3} \cdot (\mathrm{e}^{6Kt} - 1)$ and $\Im_2(r, t) \triangleq \frac{5r^2+1}{3}\mathrm{e}^{-6Kt} - \frac{2r^2+1}{3}$,

Compared with (23), we establish a self-triggered mechanism based on (24) and (25) to completely eliminate the necessity of continuously monitoring.

Theorem 1: The self-triggered mechanism is established as

$$\kappa_2(\Im_1(\|\hat{x}\|, t - t_i)) \le \mu_1\kappa_1(\Im_2(\|\hat{x}\|, t - t_i) + \mu_2) \tag{26}$$

with user-defined parameters $\mu_1, \mu_2 > 0$, $\kappa_1(\cdot) \in \mathcal{K}$ being convex and $\kappa_2(\cdot) \in \mathcal{K}_\infty$. Then, under the sample process (19) and Assumption 2, (26) excludes the Zeno behavior.

Proof: Let $r \triangleq \|\hat{x}\|$ for notational simplicity. According to the self-triggered sampling rule (26), we define the threshold function as:

$$\mathbf{T}(t; t_i) \triangleq \kappa_2(\Im_1(r, t - t_i)) - \mu_1\kappa_1(\Im_2(r, t - t_i) + \mu_2).$$

First, we show the existence and uniqueness of the solution to the transcendental equation $\mathbf{T}(t; t_i) = 0$. From the definition of $\Im_1(r, t)$ and $\Im_2(r, t)$, one can see that $\Im_1(r, t)$ and $\Im_2(r, t)$ are increasing and decreasing in t, respectively. Since the class-$\mathcal{K}$ functions κ_1, κ_2 are strictly increasing, it follows that $\mathbf{T}(t; t_i)$ is strictly increasing in t. Moreover, one can observe $\mathbf{T}(t_i; t_i) < 0$ and $\lim_{t\to\infty}\mathbf{T}(t; t_i) > 0$ because $\Im_1(r, t - t_i) \to \infty$ while $\Im_2(r, t - t_i)$ remains bounded as $t \to \infty$. Therefore, there exists a unique $t_{i+1} > t_i$ s.t. $\mathbf{T}(t_{i+1}; t_i) = 0$.

Next, the self-triggered mechanism (26) is ensured to exclude the Zeno behavior by showing that the execution interval $\Delta t_i = t_{i+1} - t_i$ is strictly positive. To this end, one can always find a function $\kappa_3(\cdot)$ s.t. $0 < \kappa_3(r) < r^2 + \mu_2$. Inspired by [25], to verify the strict positiveness of the lower bound of Δt_i, let $\kappa_3(\cdot)$ further satisfy $\Im_2(r, \Delta t_i) + \mu_2 \le \kappa_3(r) \le \Im_1(r, \Delta t_i)$, from which we obtain $\Delta t_i \geq \mathcal{T}_i > 0$, where

$$\mathcal{T}_i \triangleq \frac{1}{6K}\min\left\{ \ln\left(1+\frac{3\kappa_3(r)}{1+2r^2}\right), -\ln\left(1-\frac{3(r^2+\mu_2-\kappa_3(r))}{5r^2+1}\right)\right\}.$$

Thus, the Zeno behavior is excluded. □

Unlike ETADP designs in [17], [22], [35], the self-triggered mechanism (26) determines the next triggering instant at the current time, eliminating continuous condition checking and thereby reducing communication and computation costs.

Remark 2: The parameter K shapes the state/error bounds (e.g., see (24) and (25)) and the inter-execution interval (see Theorem 1). For resource-limited environments, K can be smaller for sparse updates while preserving system stability; for other systems, K can be larger (i.e., more sensitive and more frequent updates) while guaranteeing the Zeno-free property.

The above analysis establishes the feasibility of the self-triggered sampling rule (26). We next study the finite-gain $\mathcal{L}_2$ stability of (5) under the input $(\hat{u}^*, d^*)$.

Theorem 2: Under Assumptions 1 and 2, let the class $\mathcal{K}$ functions κ_1 and κ_2 in (26) further satisfy $\kappa_1(r) \leq r \leq \kappa_2(r)$, $\forall\, r \geq 0$. Then, the system (5) is finite-gain $\mathcal{L}_2$ stable under $(u, d) = (\hat{u}^*, d^*)$ and the self-triggered mechanism in (26).

Proof: See Appendix A. □

Remark 3: Compared with existing self-triggered designs [24], [25], [26], the sampling rule (26) offers two practical advantages. First, it avoids state-dependent denominators that may lead to Zeno behavior [24]; under Assumption 2, Zeno-free property is rigorously established in Theorem 1. Second, unlike the self-triggered schemes developed without an H_∞-oriented analysis [25], [26], (26) preserves the dissipation structure and guarantees finite-gain $\mathcal{L}_2$ stability in Theorem 2.

Remark 4: Any κ_1 and κ_2 satisfying conditions stated in Theorems 1 and 2 are admissible for the self-triggered design. Typical selections include $\kappa_1(r) = r$ and $\kappa_2(r) = r$ [25]. More generally, one may select $\kappa_1(r) = r$ and $\kappa_2(r) = r + cr^2$ with any $c > 0$.

It should be noted that the tuple (u^*, d^*) gives the Nash solution to the ZSG, but the self-triggered strategy (26) may not be the best sampling mechanism for the system state.

V. Adaptive Self-Triggered H_∞ Control

The event- and self-triggered H_∞ designs in the previous section require the tuple (u^*, d^*), which depends on the value function V^*. Since V^* is determined by the HJIE (17), which is intractable to solve directly, we integrate a STADP mechanism to approximate the Nash equilibrium (u^*, d^*) online, yielding $(\hat{u}^*, d^*)$.

In doing so, the following concept of sufficiency of excitation (SE) [41] is essential, which is a relaxation of persistency of excitation (PE) at discrete time points [32].

Definition 5: [32], [41] A bounded vector signal $\delta(t)$ is (i) PE if $\exists\, \beta > 0$ and $T > 0$ s.t. $\underline{\lambda}(\int_t^{t+T} \delta(\tau)\delta^\top(\tau)\,\mathrm{d}\,\tau) \geq \beta, \forall t \geq 0$; (ii) SE over a finite horizon if $\exists\, \beta > 0$, $T > 0$ and $t_j \in (0, T)$ for $j \in \{1, 2, \ldots, L\}$ s.t. $\underline{\lambda}(\sum_{j=1}^{L} \delta(t_j)\delta^\top(t_j)) \geqslant \beta$.

A. ACD Networks

Following the standard parameterization technique in [13], [15], [28], the value function $V^*(x)$ and the associated Nash equilibrium policies in (16) are represented in a linear-in-the-parameters form using basis functions. Specifically, the value function V^* in the HJIE (17) can be parameterized as

$$V^*(x) = \vartheta_c^\top \phi_c(x) + \varepsilon_c(x)$$

with its ideal Critic weight vector $\vartheta_c \in \mathbb{R}^{N_c}$, basis function $\phi_c(x) \in \mathbb{R}^{N_c}$ and residual error $\varepsilon_c(x) \in \mathbb{R}$; its gradient can be obtained as $\nabla V^*(x) = \nabla^\top \phi_c(x)\vartheta_c + \nabla\varepsilon_c(x)$. Accordingly, the H_∞ feedback policy u^* in (16) can be expressed using the ideal Critic network u_c^* as $u^* = u_c^*$, where

$$u_c^*(x) = -\frac{1}{2}R^{-1}g^\top(x)[\nabla^\top\phi_c(x)\vartheta_c + \nabla\varepsilon_c(x)].$$

Similarly, the ideal Actor networks $\hat{u}_a^*$ and d_a^* approximate the self-triggered control $\hat{u}^*$ and continuous disturbance d^* as $\hat{u}^*(t) = \hat{u}_a^*(t) + \varepsilon_a(\hat{x}(t))$ and $d^*(x) = d_a^*(x) + \varepsilon_d(x)$ for the residuals $\varepsilon_a(\hat{x}) \in \mathbb{R}$ and $\varepsilon_d(x) \in \mathbb{R}$, where $\hat{u}_a^*(t) \triangleq u_a^*(\hat{x}(t))$,

$$u_a^*(\hat{x}) = \vartheta_a^\top \phi_a(\hat{x}) \quad \text{and} \quad d_a^*(x) = \vartheta_d^\top \phi_d(x)$$

with their ideal weights $\vartheta_a \in \mathbb{R}^{N_u}$, $\vartheta_d \in \mathbb{R}^{N_d}$ and basis functions $\phi_a(x) \in \mathbb{R}^{N_u}$, $\phi_d(x) \in \mathbb{R}^{N_d}$. The following standard assumption is required for subsequent discussions.

Assumption 3: [28] On the compact set Ω, the basis $\phi_k(\cdot)$ and the residual $\varepsilon_k(\cdot)$ satisfy $\|\phi_k(\cdot)\| \leq \bar{\phi}_k$, $\|\nabla\phi_k(\cdot)\| \leq \bar{\phi}'_k$, $\|\varepsilon_k(\cdot)\| \leq \bar{\varepsilon}_k$ and $\|\nabla\varepsilon_k(\cdot)\| \leq \bar{\varepsilon}'_k$ with some positive constants $\bar{\phi}_k$, $\bar{\phi}'_k$, $\bar{\varepsilon}_k$, $\bar{\varepsilon}'_k$, and $k \in \{a, c, d\}$.

Remark 5: Without loss of generality, the paper considers scalar input case, where scalar input can be parameterized by weight vectors, since the POs are naturally defined for weight vectors [24], [32], [33], [39] (see Definitions 3 and 4). Meanwhile, the proposed method can be easily extended to higher-dimensional input cases by parameterizing the Actor and Disturbance estimators with weight matrices and applying the POs column-wise to each weight vector, so the same properties in Lemmas 1 and 2 remain valid.

B. Projected ACD Learning

By adopting a hybrid learning strategy, the network comprises three parts: the Critic and Disturbance are updated continuously, whereas the Actor is updated intermittently.

In what follows, we approximate the Critic ideal weight ϑ_c using its adaptive estimate $\hat{\vartheta}_c(t)$ of the Critic network

$$\hat{V}(x) \triangleq \hat{\vartheta}_c^\top \phi_c(x)$$

and define its approximation error as $\tilde{\vartheta}_c \triangleq \vartheta_c - \hat{\vartheta}_c$. Similarly, the Actor and Disturbance networks are designed as

$$u_a(\hat{x}) = \hat{\vartheta}_a^\top \phi_a(\hat{x}) \quad \text{and} \quad d_a(x) = \hat{\vartheta}_d^\top \phi_d(x), \tag{27}$$

where $\hat{\vartheta}_a$ and $\hat{\vartheta}_d$ are adaptive Actor and Disturbance weights, respectively. Define their corresponding weight errors as $\tilde{\vartheta}_a \triangleq \vartheta_a - \hat{\vartheta}_a$ and $\tilde{\vartheta}_d \triangleq \vartheta_d - \hat{\vartheta}_d$.

To avoid confusion, signals evaluated at the sampled state $\hat{x}$ (e.g., $u_a(\hat{x})$, $\phi_a(\hat{x})$, $u^*(\hat{x})$) are denoted by $\hat{u}_a$, $\hat{\phi}_a$, and $\hat{u}^*$, respectively. Signals using continuous state feedback (e.g., $d_a(x)$,

$\phi_d(x)$, $d^*(x)$) are written as d_a, ϕ_d, and d^*. The implemented input pair $(\hat{u}_a, d_a)$ serves as an estimate of the Nash equilibrium $(\hat{u}^*, d^*)$ for (5).

1) Continuous Critic Learning: The first step is to approximate the Hamiltonian $\mathcal{H}(\cdot)$ in the HJIE (17) by

$$\mathcal{H}(x, \hat{u}_a, d_a, \nabla\hat{V}) = \nabla^\top \hat{V}(x) F(x, \hat{u}_a, d_a) + \ell(x, \hat{u}_a, d_a)$$

which then defines the Critic error as

$$\begin{aligned} e_c &\triangleq \mathcal{H}(x, \hat{u}_a, d_a, \nabla\hat{V}) - \mathcal{H}(x, u^*, d^*, \nabla V^*) \\ &\equiv \hat{\vartheta}_c^\top(t)\sigma_c(t) + \ell(x(t), \hat{u}_a(t), d_a(t)), \end{aligned} \tag{28}$$

where $\sigma_c \triangleq \nabla\phi_c \cdot F(x, \hat{u}_a, d_a)$. To relax the PE condition, define the composite Critic error w.r.t. the data at past instants $\mathfrak{c}_j < t$ for $j = 1, \dots, L_c$ as

$$e_c(\mathfrak{c}_j, t) \triangleq \hat{\vartheta}_c^\top(t)\sigma_c(\mathfrak{c}_j) + \ell(x(\mathfrak{c}_j), \hat{u}_a(\mathfrak{c}_j), d_a(\mathfrak{c}_j)).$$

where both the past data $\sigma_c(\mathfrak{c}_j)$ and the current weight $\hat{\vartheta}_c(t)$ are involved. For notational simplicity, let $e_{\mathfrak{c}_j}(t) = e_c(\mathfrak{c}_j, t)$, $\sigma_{\mathfrak{c}_j} = \sigma_c(\mathfrak{c}_j)$, and $\ell_{\mathfrak{c}_j} = \ell(\mathfrak{c}_j)$. Then, one has $e_{\mathfrak{c}_j}(t) = \hat{\vartheta}_c^\top(t)\sigma_{\mathfrak{c}_j} + \ell_{\mathfrak{c}_j}$, whereas $e_c(t) = \hat{\vartheta}_c^\top(t)\sigma_c(t) + \ell(t)$.

Let the objective E_c for Critic learning be defined as the normalized composite error

$$E_c(\tilde{\vartheta}_c) \triangleq \frac{1}{2} \cdot \left(\frac{e_c^2}{(1 + \|\sigma_c\|^2)^2} + \sum_{j=1}^{L_c} \frac{e_{\mathfrak{c}_j}^2}{(1 + \|\sigma_{\mathfrak{c}_j}\|^2)^2} \right),$$

and we propose the Critic learning for minimizing E_c

$$\dot{\hat{\vartheta}}_c = -\alpha_c \left(\frac{\sigma_c e_c}{(1 + \|\sigma_c\|^2)^2} + \sum_{j=1}^{L_c} \frac{\sigma_{\mathfrak{c}_j} e_{\mathfrak{c}_j}}{(1 + \|\sigma_{\mathfrak{c}_j}\|^2)^2} \right). \tag{29}$$

2) Intermittent Actor Learning: From the Critic components, we define the Actor's target $u_c \triangleq -\frac{1}{2}R^{-1}g^\top\nabla^\top\phi_c\hat{\vartheta}_c$ and the adaptation error $e_a \triangleq \hat{u}_a - u_c$, i.e.,

$$e_a = \hat{\vartheta}_a^\top\hat{\phi}_a + \tfrac{1}{2}R^{-1}g^\top\nabla^\top\phi_c\hat{\vartheta}_c = -\tilde{\vartheta}_a^\top\hat{\phi}_a + \epsilon_a, \tag{30}$$

where $\epsilon_a \triangleq \vartheta_a^\top\hat{\phi}_a + \frac{1}{2}R^{-1}g^\top\nabla^\top\phi_c\hat{\vartheta}_c$. Employing the similar notations in Critic learning (29), denote $\hat{\phi}_{\mathfrak{a}_j}$ as the basis $\hat{\phi}_a(\mathfrak{a}_j)$ sampled at past instants $\mathfrak{a}_j < t$ for $j = 1, \dots, L_a$, and define the composite Actor error at those instants as $e_{\mathfrak{a}_j} \triangleq -\tilde{\vartheta}_a^\top\hat{\phi}_{\mathfrak{a}_j} + \epsilon_{\mathfrak{a}_j}$, where $\epsilon_{\mathfrak{a}_j} \triangleq \vartheta_a^\top\hat{\phi}_{\mathfrak{a}_j} + \frac{1}{2}R^{-1}(g^\top\nabla^\top\phi_c\hat{\vartheta}_c)(\mathfrak{a}_j)$ and all the signals except $\hat{\vartheta}_a(t)$ are sampled at time $\mathfrak{a}_j < t$.

The Actor learning aims to minimize the objective function

$$E_a(\tilde{\vartheta}_a) \triangleq \frac{1}{2}\left(e_a^2 + \textstyle\sum_{j=1}^{L_a} e_{\mathfrak{a}_j}^2\right).$$

Inspired by the PO and its property as in Lemma 2, we design the following Actor tuning law:

$$\begin{aligned} &\dot{\hat{\vartheta}}_a(t) = 0 \qquad \forall t \in [t_i, t_{i+1}), \\ &\hat{\vartheta}_a^+(t) = \mathrm{Proj}_{\mathrm{II}}(\widehat{\vartheta}_a(t)\,|\psi_a) \; \forall t = t_{i+1}, \end{aligned} \tag{31}$$

where $\psi_a > 0$ and $\widehat{\vartheta}_a \triangleq \hat{\vartheta}_a - \alpha_a\nabla E_a$, with a sufficiently small learning rate $\alpha_a > 0$ and $\nabla E_a = \frac{\partial E_a}{\partial\hat{\vartheta}_a} = \hat{\phi}_a e_a + \sum_{j=1}^{L_a}\hat{\phi}_{\mathfrak{a}_j}e_{\mathfrak{a}_j}$. By Lemma 2, the Actor weight $\hat{\vartheta}_a$ is bounded as $\|\hat{\vartheta}_a\| \leq \psi_a$.

3) Continuous Disturbance Learning: With the Critic network again, define Disturbance's target $d_c \triangleq \frac{1}{2\gamma^2}k^\top\nabla^\top\phi_c\hat{\vartheta}_c$ and the Disturbance learning error $e_d \triangleq d_a - d_c$, i.e.,

$$e_d = \hat{\vartheta}_d^\top\phi_d - \frac{1}{2\gamma^2}k^\top\nabla^\top\phi_c\hat{\vartheta}_c = -\tilde{\vartheta}_d^\top\phi_d + \epsilon_d \tag{32}$$

where $\epsilon_d \triangleq \vartheta_d^\top\phi_d - \frac{1}{2\gamma^2}k^\top\nabla^\top\phi_c\hat{\vartheta}_c$. Similarly to the Actor and Critic learning schemes above, to weaken the PE condition, we collect the data at past instants $\mathfrak{d}_j < t$ for $j = 1, \dots, L_d$ and construct the Disturbance learning error as $e_{\mathfrak{d}_j} \triangleq -\tilde{\vartheta}_d^\top\phi_{\mathfrak{d}_j} + \epsilon_{\mathfrak{d}_j}$, where $\epsilon_{\mathfrak{d}_j} \triangleq \vartheta_d^\top\phi_{\mathfrak{d}_j} - \frac{1}{2\gamma^2}(k^\top\nabla^\top\phi_c\hat{\vartheta}_c)(\mathfrak{d}_j)$ and $\phi_{\mathfrak{d}_j} \triangleq \phi_d(\mathfrak{d}_j)$.

The Disturbance learning aims to minimize:

$$E_d(\tilde{\vartheta}_d) \triangleq \frac{1}{2}\left(e_d^2 + \textstyle\sum_{j=1}^{L_d} e_{\mathfrak{d}_j}^2\right),$$

and we design the projected gradient descent law as

$$\dot{\hat{\vartheta}}_d = \mathrm{Proj}_{\mathrm{I}}(-\alpha_d\nabla E_d, \hat{\vartheta}_d\,|\mathcal{G}, \Omega_{\mathcal{G}}, \alpha_d), \tag{33}$$

where $\mathcal{G}(\hat{\vartheta}_d) = \frac{1}{2}(\|\hat{\vartheta}_d\|^2 - \psi_d^2)$ with the constraint $\psi_d > 0$, and $\nabla E_d = \frac{\partial E_d}{\partial\hat{\vartheta}_d} = \phi_d e_d + \sum_{j=1}^{L_d}\phi_{\mathfrak{d}_j}e_{\mathfrak{d}_j}$. By Lemma 1, the adaptive weight $\hat{\vartheta}_d$ is bounded as $\|\hat{\vartheta}_d\| \leq \psi_d$.

For the ACD learning, the Critic learning rate α_c should dominate those of the Actor and Disturbance estimators, to maintain stability, accurate learning, and fast convergence. In the H_∞ control sense, α_a should dominate α_d to enhance the robustness (disturbance attenuation). Meanwhile, as pointed out by [32], [33], the learning rates α_a and α_d should be sufficiently small to guarantee weight convergence.

Remark 6: As summarized in Table I, this work targets a STADP implementation to eliminate continuous monitoring in ETC [20], [22] while achieving H_∞ performance. Although some ADP schemes can be implemented with unknown drift dynamics f, the input dynamics g and k remain necessary for Actor/Disturbance learning, since the update laws contain the learning errors that explicitly depend on g and k (see (30) and (33)), respectively. Nevertheless, the proposed framework is readily extendable to unknown drift dynamics via an identifier [22], [39] or data-driven integral reinforcement learning [20], [40] for iteratively solving the HJIE (17).

C. Stability and Convergence Analysis

Next, we analyze the closed-loop stability of (5) under the ACD learning architecture and the self-triggered sampling rule, with emphasis on boundedness and convergence, using a Lyapunov-based approach.

1) Flow Dynamics of State $x(t)$: With the online Actor and Disturbance in (27), taking the time derivative of $V^*(x)$ along the system (5) and following the similar procedure described in the proof of Theorem 2 (see Appendix A), we obtain

$$\dot{V}^* \leq \bar{\lambda}(R)\|u^* - \hat{u}_a\|^2 - \|\hat{u}_a\|_R^2 - \gamma^2(\|d^* - d_a\|^2 - \|d_a\|^2) - Q,$$

where the terms $\|d_a\|^2$ and $\|u^* - \hat{u}_a\|^2$ can be bounded as $\|d_a\|^2 \leq \psi_d^2\bar{\phi}_d^2$ and

$$\begin{aligned} \|u^* - \hat{u}_a\|^2 &\leq 2\|u^* - \hat{u}^*\|^2 + 2\|\hat{u}^* - \hat{u}_a\|^2 \\ &\leq 2\mathcal{L}_u^2\|\tilde{x}\|^2 + 4\bar{\phi}_a^2(\|\vartheta_a\| + \psi_a)^2 + 4\bar{\varepsilon}_a^2, \end{aligned}$$

by (18) and Assumption 3, and the bounds $\|\hat{\vartheta}_a\| \leq \psi_a$ and $\|\hat{\vartheta}_d\| \leq \psi_d$, thanks to the projections.

Substituting the above bounds into $\dot{V}^*(x)$ yields

$$\dot{V}^*(x) \leq 2\mathcal{L}_u^2\bar{\lambda}(R)\|\tilde{x}\|^2 - q_1\bar{\alpha}\|x\|^2 - q_1\alpha\|x\|^2 - \|\hat{u}_a\|_R^2 + \omega$$

for $\omega \triangleq 4\bar{\lambda}(R)\bar{\phi}_a^2(\|\vartheta_a\| + \psi_a)^2 + \gamma^2\psi_d^2\bar{\phi}_d^2 + 4\bar{\lambda}(R)\bar{\varepsilon}_a^2$, constant $\alpha \in (0,1)$ and its conjugate $\bar{\alpha} \triangleq 1-\alpha$.

Let μ_1 and μ_2 in (26) be chosen s.t.

$$\mu_1 = \frac{q_1\bar{\alpha}}{2\mathcal{L}_u^2\bar{\lambda}(R)}, \qquad \frac{\mu_1}{2}\kappa_1(2\mu_2) = \frac{\|\hat{u}_a\|_R^2 + \zeta}{2\mathcal{L}_u^2\bar{\lambda}(R)} \tag{34}$$

for $\zeta \in (0, \omega)$.

Then, following the similar procedure to that in the proof of Theorem 2, one has $2\mathcal{L}_u^2\bar{\lambda}(R)\|\tilde{x}\|^2 - q_1\bar{\alpha}\|x\|^2 - \bar{\lambda}(R)\|u_a\|^2 - \zeta \leq 0$.

Hence, $\dot{V}^*(x)$ can be further expressed with $\omega_0 \triangleq \omega - \zeta$ as

$$\dot{V}^*(x) \leq -q_1\alpha\|x\|^2 + \omega_0. \tag{35}$$

2) Jump Dynamics of Sampled State $\hat{x}$: From (19), the discrete dynamics of the sampled state $\hat{x}^+(t)$ is obtained as

$$\begin{aligned} \dot{\hat{x}}(t) &= 0 \qquad \text{for all } t \in [t_i, t_{i+1}), \\ \hat{x}^+(t) &= x(t) \quad \text{for each } t = t_{i+1} \end{aligned} \tag{36}$$

Recall c_1, c_2 in Assumption 1. Then, $\dot{V}^* \leq -q_1\alpha V^*/c_2 + \omega_0$ from (36) and by the comparison Lemma [37],

$$V^*(x(t)) \leq (V^*(x(t_i)) - \nu) \cdot \mathrm{e}^{-\varsigma(t-t_i)} + \nu \quad \forall t \in [t_i, t_{i+1}),$$

where ς and ν denote $q_1\alpha/c_2$ and $\omega_0\varsigma^{-1}$, respectively.

At the triggering instant $t = t_{i+1}$, V^* further satisfies

$$V^*(x(t_{i+1})) \leq (V^*(x(t_i)) - \nu) \cdot \mathrm{e}^{-\varsigma\Delta t_i} + \nu$$

for $\Delta t_i \triangleq t_{i+1} - t_i$, which can be equivalently expressed as

$$\Delta V^*(\hat{x}) \leq -\eta_a V^*(\hat{x}) + \eta_b \leq -c_1\eta_a\|\hat{x}\|^2 + \eta_b \tag{37}$$

with $\Delta V^*(\hat{x}) \triangleq V^*(\hat{x}^+) - V^*(\hat{x}), \eta_a \triangleq 1 - \mathrm{e}^{-\varsigma\Delta t_i}, \eta_b \triangleq \nu\eta_a$.

3) Flow Dynamics of Critic Weight $\hat{\vartheta}_c$: By Lemma 3 and (29), the Critic error e_c given by (28) can be rewritten as

$$e_c = \mathcal{H}(x, \hat{u}_a, d_a, \nabla\hat{V}) - \mathcal{H}(x, \hat{u}^*, d^*, \nabla V^*) + \|u^* - \hat{u}^*\|_R^2,$$

where each term of the Hamiltonian

$$\mathcal{H}(\cdot, \hat{u}^*, d^*, \nabla V^*) = \nabla^\top V^* F(\cdot, \hat{u}^*, d^*) + \ell(\cdot, \hat{u}^*, d^*)$$

can be expanded with $\tilde{u}_a \triangleq \hat{u}^* - \hat{u}_a$ and $\tilde{d}_a \triangleq d_a - d^*$ as

$$\begin{aligned} \nabla^\top V^* F(\cdot, \hat{u}^*, d^*) &= \left(\vartheta_c^\top \nabla\phi_c\right) F(\cdot, \hat{u}^*, d^*) + \nabla^\top\varepsilon_c F(\cdot, \hat{u}^*, d^*) \\ &= \vartheta_c^\top\sigma_c + \underbrace{\vartheta_c^\top\nabla\phi_c(g\tilde{u}_a + k\tilde{d}_a) + \nabla^\top\varepsilon_c F(\cdot, \hat{u}^*, d^*)}_{\triangleq\epsilon_1} \end{aligned}$$

$$\begin{aligned} \ell(\cdot, \hat{u}^*, d^*) &= Q + \|\hat{u}_a + \tilde{u}_a\|_R^2 - \gamma^2\|d_a + \tilde{d}_a\|^2 \\ &= \ell(x, \hat{u}_a, d_a) + \underbrace{2\hat{u}_a^\top R\tilde{u}_a + \|\tilde{u}_a\|_R^2 - \gamma^2(2d_a^\top\tilde{d}_a + \|\tilde{d}_a\|^2)}_{\triangleq\epsilon_2} \end{aligned}$$

Hence, the Critic errors e_c and $e_{\mathfrak{c}_j}$ can be written as

$$e_c = -\tilde{\vartheta}_c^\top\sigma_c - \epsilon_c \quad \text{and} \quad e_{\mathfrak{c}_j} = -\tilde{\vartheta}_c^\top\sigma_{\mathfrak{c}_j} - \epsilon_{\mathfrak{c}_j} \tag{38}$$

with their residuals $\epsilon_c \triangleq \epsilon_1 + \epsilon_2 + \epsilon_3$ and $\epsilon_{\mathfrak{c}_j} \triangleq \epsilon_c(\mathfrak{c}_j)$, where ϵ_1 and ϵ_2 are defined above, and $\epsilon_3 \triangleq -\|u^* - \hat{u}^*\|_R^2$. Here,

$$\hat{u}_a = \hat{\vartheta}_a^\top\hat{\phi}_a,\ \hat{u}^* = \vartheta_a^\top\hat{\phi}_a + \hat{\varepsilon}_a,\ d_a = \hat{\vartheta}_d^\top\phi_d,\ d^* = \vartheta_d^\top\phi_d + \varepsilon_d,$$

(which are true by definitions, e.g., see (27)) are bounded as

$$\begin{aligned} \|\hat{u}_a\| &\leq \psi_a\bar{\phi}_a \triangleq \hat{U}_a \qquad \|\hat{u}^*\| \leq \|\vartheta_a\|\bar{\phi}_a + \bar{\varepsilon}_a \triangleq \hat{U}^* \\ \|d_a\| &\leq \psi_d\phi_d \triangleq D_a \qquad \|d^*\| \leq \|\vartheta_d\|\bar{\phi}_d + \bar{\varepsilon}_d \triangleq D^* \end{aligned}$$

by Assumption 3 and the bounds $\|\hat{\vartheta}_a\| \leq \psi_a$ and $\|\hat{\vartheta}_d\| \leq \psi_d$.

Similarly, the terms $\tilde{u}_a$ and $\tilde{d}_a$ are bounded as

$$\begin{aligned} \|\tilde{u}_a\| &\leq (\psi_a + \|\vartheta_a\|)\bar{\phi}_a + \bar{\varepsilon}_a \triangleq \tilde{U}_a \\ \|\tilde{d}_a\| &\leq (\psi_d + \|\vartheta_d\|)\bar{\phi}_d + \bar{\varepsilon}_d \triangleq \tilde{D}_a \end{aligned}$$

and thus, with Assumption 3, ϵ_1 and ϵ_2 are bounded as

$$\epsilon_1 \leq \underbrace{\|\vartheta_c\|\bar{\phi}_c'(b_g\tilde{U}_a + b_k\tilde{D}_a) + \bar{\varepsilon}_c'(b_g\hat{U}^* + b_kD^*)}_{\triangleq\epsilon_{1,\max}} + \bar{\varepsilon}_c'b_f\|x\|$$

$$\epsilon_2 \leq \bar{\lambda}(R)(2\hat{U}_a\tilde{U}_a + \tilde{U}_a^2) + \gamma^2(2D_a\tilde{D}_a + \tilde{D}_a^2) \triangleq \epsilon_{2,\max}$$

and $\epsilon_3 = -\|u^* - \hat{u}^*\|_R^2 = -\|\vartheta_a^\top(\phi_a - \hat{\phi}_a) + (\varepsilon_a - \hat{\varepsilon}_a)\|_R^2$ as

$$\epsilon_3 \leq \bar{\lambda}(R)\left(2\|\vartheta_a\|\bar{\phi}_a + 2\bar{\varepsilon}_a\right)^2 \triangleq \epsilon_{3,\max}. \tag{39}$$

Denoting $\bar{\sigma}_c \triangleq \sigma_c/(1 + \|\sigma_c\|^2)$ and $\bar{\sigma}_{\mathfrak{c}_j} \triangleq \sigma_{\mathfrak{c}_j}/(1 + \|\sigma_{\mathfrak{c}_j}\|^2)$, and substituting the Critic errors (38) into (29), we obtain

$$\dot{\tilde{\vartheta}}_c = -\alpha_c\left[\left(\bar{\sigma}_c\bar{\sigma}_c^\top + \mathfrak{C}_1\right)\tilde{\vartheta}_c + \mathfrak{C}_2\right] \tag{40}$$

where $\mathfrak{C}_1 \triangleq \sum_{j=1}^{L_c}\bar{\sigma}_{\mathfrak{c}_j}\bar{\sigma}_{\mathfrak{c}_j}^\top$ and $\mathfrak{C}_2 \triangleq \frac{\sigma_c\epsilon_c}{(1+\|\sigma_c\|^2)^2} + \sum_{j=1}^{L_c}\frac{\sigma_{\mathfrak{c}_j}\epsilon_{\mathfrak{c}_j}}{(1+\|\sigma_{\mathfrak{c}_j}\|^2)^2}$.

Now, consider $V_c(\tilde{\vartheta}_c) \triangleq \frac{1}{2}\tilde{\vartheta}_c^\top\alpha_c^{-1}\tilde{\vartheta}_c$ as a Lyapunov candidate. Then, differentiating it along the dynamics (40) yields

$$\dot{V}_c(\tilde{\vartheta}_c) = -\tilde{\vartheta}_c^\top\left(\bar{\sigma}_c\bar{\sigma}_c^\top - \mathfrak{C}_1\right)\tilde{\vartheta}_c + \tilde{\vartheta}_c^\top\mathfrak{C}_2. \tag{41}$$

4) Jump Dynamics of Actor Weight $\hat{\vartheta}_a$: The error dynamics of the intermittent Actor weight learning (31) is as follows,

$$\begin{aligned} \dot{\tilde{\vartheta}}_a(t) &= 0, \qquad \forall t \in [t_i, t_{i+1}), \\ \tilde{\vartheta}_a^+(t) &= \widehat{\vartheta}_a^*(t) - \mathrm{Proj}_{\Pi}(\widehat{\vartheta}_a(t)\,|\,\psi_a) \ \ \forall t = t_{i+1}, \end{aligned} \tag{42}$$

where the term $\widehat{\vartheta}_a^* = \vartheta_a^* - \alpha_a\hat{\phi}_a(\hat{u}^* - u^*)$. With (31), we further have at each sampling instant t_{i+1}

$$\tilde{\vartheta}_a^+ = \tilde{\vartheta}_a - \alpha_a[(\mathfrak{A}_0 + \mathfrak{A}_1)\tilde{\vartheta}_a - \mathfrak{A}_2] - \omega_1(\tfrac{1}{2}\omega_2 + \hat{\varepsilon}_a) - p \tag{43}$$

where p, ω_i $(i = 1, 2)$ and $\mathfrak{A}_i$ $(i = 0, 1, 2)$ are defined as

$$\begin{aligned} &p \triangleq \mathbf{1}\{\psi_a < \|\widehat{\vartheta}_a\|\} \cdot (\psi_a - \|\widehat{\vartheta}_a\|) \cdot \widehat{\vartheta}_a/\|\widehat{\vartheta}_a\|, \\ &\omega_1 \triangleq \alpha_a\hat{\phi}_a, \quad \omega_2 \triangleq R^{-1}g^\top(\nabla^\top\phi_c\hat{\vartheta}_c + \nabla^\top\varepsilon_c), \\ &\mathfrak{A}_0 \triangleq \hat{\phi}_a\hat{\phi}_a^\top, \quad \mathfrak{A}_1 \triangleq \textstyle\sum_{j=1}^{L_a}\hat{\phi}_{\mathfrak{a}_j}\hat{\phi}_{\mathfrak{a}_j}^\top, \\ &\mathfrak{A}_2 \triangleq \textstyle\sum_{j=1}^{L_a}(\hat{\phi}_{\mathfrak{a}_j}\hat{\phi}_{\mathfrak{a}_j}^\top\vartheta_a + \dfrac{1}{2}\hat{\phi}_aR^{-1}g^\top\nabla^\top\phi_c\hat{\vartheta}_c). \end{aligned} \tag{44}$$

Consider $V_a(\tilde{\vartheta}_a) = \|\tilde{\vartheta}_a\|^2$ as a Lyapunov function for the Actor learning error, whose difference $\Delta V_a(\tilde{\vartheta}_a) \triangleq V_a(\tilde{\vartheta}_a^+) - V_a(\tilde{\vartheta}_a) = \|\tilde{\vartheta}_a^+\|^2 - \|\tilde{\vartheta}_a\|^2$ can be expressed as:

$$\begin{aligned}&= \|p\|^2 - 2p^\top(\widehat{\vartheta}_a^* - \widehat{\vartheta}_a) + \alpha_a^2\tilde{\vartheta}_a^\top(\mathfrak{A}_0 + \mathfrak{A}_1)^2\tilde{\vartheta}_a \\ &\quad + \kappa + \tilde{\vartheta}_a^\top\varpi - 2\alpha_a\tilde{\vartheta}_a^\top(\mathfrak{A}_0 + \mathfrak{A}_1)\tilde{\vartheta}_a \end{aligned} \tag{45}$$

where the terms κ and ϖ are defined as

$$\begin{aligned}\kappa \triangleq\ & \|\omega_1\|^2\omega_2^2/4 + \|\omega_1\|^2\hat{\varepsilon}_a^2 - \omega_1^\top\omega_2\alpha_a\mathfrak{A}_2 \\ & + \|\omega_1\|^2\omega_2\hat{\varepsilon}_a + \alpha_a^2\|\mathfrak{A}_2\|^2 - 2\alpha_a\varepsilon_a\omega_1^\top\mathfrak{A}_2, \\ \varpi \triangleq\ & -\omega_1\omega_2 + \alpha_a(\mathfrak{A}_0 + \mathfrak{A}_1)\omega_1\omega_2 + 2\alpha_a(\mathfrak{A}_0 + \mathfrak{A}_1)\omega_1\hat{\varepsilon}_a \\ & - 2\alpha_a^2(\mathfrak{A}_0 + \mathfrak{A}_1)\mathfrak{A}_2 - 2\omega_1\hat{\varepsilon}_a + 2\alpha_a\mathfrak{A}_2.\end{aligned}$$

5) Flow Dynamics of Disturbance Weight $\hat{\vartheta}_d$: Consider $V_d(\tilde{\vartheta}_d) = \frac{1}{2}\tilde{\vartheta}_d^\top\alpha_d^{-1}\tilde{\vartheta}_d$ as a Lyapunov function for Disturbance. By (33), the dynamics of $\tilde{\vartheta}_d$ can be formulated as

$$\dot{\tilde{\vartheta}}_d = -\alpha_d\left(\phi_d\phi_d^\top + \mathfrak{D}_1\right)\tilde{\vartheta}_d + \alpha_d\mathfrak{D}_1, \tag{46}$$

where $\mathfrak{D}_1$ and $\mathfrak{D}_2$ are defined as $\mathfrak{D}_1 \triangleq \sum_{j=1}^{L_d}\phi_{\mathfrak{d}_j}\phi_{\mathfrak{d}_j}^\top$ and

$$\mathfrak{D}_2 \triangleq \phi_d\epsilon_d + \sum_{l=1}^{L_d}\phi_{\mathfrak{d}_j}\epsilon_{\mathfrak{d}_j} + \mathfrak{D}_3\left[\phi_d e_d + \sum_{j=1}^{L_d}\phi_{\mathfrak{d}_j}e_{\mathfrak{d}_j}\right] \tag{47}$$

with $\mathfrak{D}_3 \triangleq \mathbf{1}\{\hat{\vartheta}_d \in \Omega\} \cdot \nabla\mathcal{G}\nabla^\top\mathcal{G}/\|\nabla\mathcal{G}\|^2$.

Differentiating $V_d(\tilde{\vartheta}_d(t))$ along (46) yields

$$\dot{V}_d(\tilde{\vartheta}_d) = -\tilde{\vartheta}_d^\top\left(\phi_d\phi_d^\top + \mathfrak{D}_1\right)\tilde{\vartheta}_d + \tilde{\vartheta}_d^\top\mathfrak{D}_2. \tag{48}$$

6) Overall Stability and Convergence: The theoretical discussions so far can be summarized by the next theorem.

Theorem 3: Consider the system (5) under $(u, d) = (\hat{u}_a, d_a)$ with the ACD learning laws (29), (31), (33) and the self-triggered mechanism in (26). Then, under Assumptions 1 to 3, the augmented state $\mathcal{X} \triangleq [x^\top, \hat{x}^\top, \tilde{\vartheta}_c^\top, \tilde{\vartheta}_a^\top, \tilde{\vartheta}_d^\top]^\top$ is uniformly ultimately bounded (UUB), provided that the signals $\bar{\sigma}_c$, ϕ_d, $\hat{\phi}_a$, with the sample points $\{\bar{\sigma}_{\mathfrak{c}_j}\}_{j=1}^{L_c}$, $\{\phi_{\mathfrak{d}_j}\}_{j=1}^{L_d}$, $\{\hat{\phi}_{\mathfrak{a}_j}\}_{j=1}^{L_a}$, being SE.

Proof: See Appendix B. □

Theorem 4: Under the conditions in Theorem 3, (i) the Critic error in (28) is UUB; (ii) the tuple $(\hat{u}_a, d_a)$ converges to a neighborhood of the Nash solution (16) to the ZSG (14).

Proof: 1) The Critic error (28) can be bounded with the related bounds established above and $\epsilon_{\max} \triangleq \sum_{i=1}^3 \epsilon_{i,\max}$ as

$$\begin{aligned}|e_c| &\le \|\tilde{\vartheta}_c\| \cdot \|\nabla^\top\phi_c \cdot F(x, \hat{u}_a, u_d)\| + \textstyle\sum_{i=1}^3\|\epsilon_i\| \\ &\le \|\tilde{\vartheta}_c\| \cdot \bar{\phi}_c' \cdot \left(b_f\|x\| + b_g\hat{U}_a + b_k U_d\right) + (\epsilon_{\max} + \bar{\varepsilon}_c' b_f\|x\|),\end{aligned}$$

where all of the terms except for $\tilde{\vartheta}_c$ and x are constants. Thus, e_c is UUB since so are x and $\tilde{\vartheta}_c$ by Theorem 3.

2) From (27) and Assumption 3, the following hold

$$\begin{aligned}\|\hat{u}_a - \hat{u}^*\| &= \|\tilde{\vartheta}_a^\top\hat{\phi}_a - \hat{\varepsilon}_a\| \le \bar{\phi}_a\|\tilde{\vartheta}_a\| + \bar{\varepsilon}_a \\ \|d_a - d^*\| &= \|\tilde{\vartheta}_d^\top\hat{\phi}_d - \varepsilon_d\| \le \bar{\phi}_d\|\tilde{\vartheta}_d\| + \bar{\varepsilon}_d\end{aligned}$$

Since both $\tilde{\vartheta}_a$ and $\tilde{\vartheta}_d$ are UUB, one can conclude that $(\hat{u}_a, d_a)$ converges to a neighborhood of (u^*, d^*). □

D. Computational Complexity Analysis for Projection

As summarized in Table I, POs are widely adopted in ADP to enforce constraints of adaptive weights in both optimal control [20], [24], [39] and game-theoretic problems [34]. However, the literature rarely investigates whether POs would bring extra computational complexity in ADP learning, which motivates this subsection.

Since the objective functions E_a and E_d for the Actors and Disturbance updates are scalar and quadratic in weight vectors $\hat{\vartheta}_a$ and $\hat{\vartheta}_d$ (see Subsections V-B2 and V-B3), we omit the composite errors in E_a and E_d without loss of generality. Then, the minimization of E_a and E_d can be simplified into

$$\min\mathcal{J}_d(\hat{\vartheta}_d), \quad \mathcal{J}_d(\hat{\vartheta}_d) \triangleq \frac{1}{2}(\hat{\vartheta}_d^\top\phi_d - d_c)^2, \tag{49}$$

$$\min\mathcal{J}_a(\hat{\vartheta}_a), \quad \mathcal{J}_a(\hat{\vartheta}_a) \triangleq \frac{1}{2}(\hat{\vartheta}_a^\top\hat{\phi}_a - u_c)^2. \tag{50}$$

For (49), the gradient $\nabla\mathcal{J}_d(\hat{\vartheta}_d) = (\hat{\vartheta}_d^\top\phi_d - d_c)\phi_d$ is dominated by one inner product and one scalar-vector multiplication, requiring time complexity $\mathcal{O}(N_d)$ and $3N_d$ flops. Throughout this paper, the flop count includes only basic floating-point arithmetic; operations such as $\max\{\cdot,\cdot\}$, $\mathrm{sign}(\cdot)$, and $\sqrt{\cdot}$ are excluded.

The Type-I-projected learning can be expressed as

$$\dot{\hat{\vartheta}}_d = -\alpha_d\nabla\mathcal{J}_d(\hat{\vartheta}_d) + m_{d,1}\cdot m_{d,2}\cdot\frac{\hat{\vartheta}_d\hat{\vartheta}_d^\top}{\hat{\vartheta}_d^\top\hat{\vartheta}_d}\nabla\mathcal{J}_d(\hat{\vartheta}_d), \tag{51}$$

where $m_{d,1} \triangleq 1 - \mathrm{sign}(\|\hat{\vartheta}_d\| - \psi_d)$ and $m_{d,2} \triangleq \max\{0, \mathrm{sign}((-\alpha_d\nabla\mathcal{J}_d(\hat{\vartheta}_d))^\top\hat{\vartheta}_d)\}$. Then, the unprojected case $\dot{\hat{\vartheta}}_d = -\alpha_d\nabla\mathcal{J}_d(\hat{\vartheta}_d)$ requires time complexity $\mathcal{O}(N_d)$ and $4N_d$ flops. When the Type-I PO is activated, computing the second term in (51) preserves the same order-wise complexity $\mathcal{O}(N_d)$ but requires extra $7N_d + 3$ flops.

For (50), the (discrete-time) gradient step

$$\hat{\vartheta}_a^+ = \hat{\vartheta}_a - \alpha_a\nabla\mathcal{J}_a(\hat{\vartheta}_a), \quad \nabla\mathcal{J}_a(\hat{\vartheta}_a) = (\hat{\vartheta}_a^\top\hat{\phi}_a - u_c)\hat{\phi}_a$$

has complexity $\mathcal{O}(N_u)$, requiring $5N_u$ flops.

Considering PO, we rewrite the Type-II (see Definitions 4) projected gradient descent in an explicit form as

$$\hat{\vartheta}_a^+ = \widehat{\vartheta}_a + \max\{\mathrm{sign}(\|\widehat{\vartheta}_a\| - \psi_a), 0\}\cdot\frac{\psi_a - \|\widehat{\vartheta}_a\|}{\|\widehat{\vartheta}_a\|}\cdot\widehat{\vartheta}_a, \tag{52}$$

where $\widehat{\vartheta}_a = \hat{\vartheta}_a - \alpha_a\nabla\mathcal{J}_a(\hat{\vartheta}_a)$. This only introduces norm computations and scalar–vector multiplication, and thus does not change the order-wise complexity. But obtaining the second term in (52) brings additional $4N_u + 2$ flops.

Collecting the above discussions and as summarized in Table II, although the POs increase the update flops, it does not change the order-wise computational complexity. More importantly, PO enforces boundedness of the adaptive weights, which is a key property for guaranteeing stability of the learning dynamics. Hence, the additional flops constitute a modest and practically acceptable overhead in exchange for a stability-guaranteeing safeguard.

TABLE II
COMPARISON OF TIME COMPLEXITY AND FLOPS

		Unprojected	Projected
Actor Update	Complexity	$\mathcal{O}(N_u)$	$\mathcal{O}(N_u)$
	Flops	$5N_u$	$9N_u+2$
Disturbance Update	Complexity	$\mathcal{O}(N_d)$	$\mathcal{O}(N_d)$
	Flops	$4N_d$	$11N_d+3$

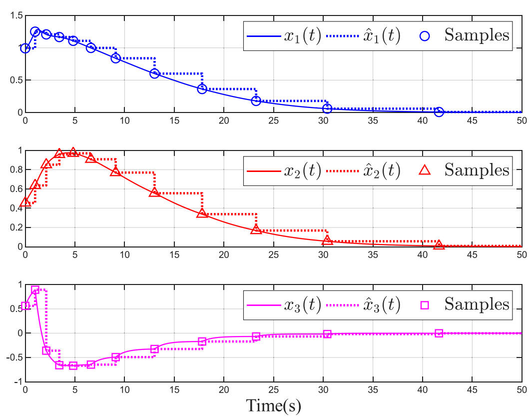

Fig. 2. Evolution of the system state with its samples.

VI. SIMULATION

In this section, two practical examples are presented to verify the effectiveness of the projected STADP algorithm by comparisons with ETC and TTC. The self-triggered rule (26) is determined by $K = 0.1$, $\kappa_1(r) = r$ and $\kappa_2(r) = r$, s.t. $\Im_1(\|\hat{x}\|, t - t_i) \leq \mu_1 \Im_2(\|\hat{x}\|, t - t_i) + \mu_1\mu_2$, where μ_1 and μ_2 being defined in (34) with $\bar{\alpha} = 0.5$, $\zeta = 0.1$ and $\mathcal{L}_u = 0.2$.

A. Case 1: F16 Aircraft Plant

Following the continuous-time F-16 plant adopted in [42] (with a quadratic cost s.t. $Q = I_2$, $R = 1$ and $\gamma = 5$), Let the state $x = [\omega_a, \omega_p, \omega_e]^\top$ collect the angle of attack ω_a, pitch rate ω_p, and elevator deflection ω_e. The control signal u is the elevator actuator voltage, while the input d represents wind-gust disturbance acting on the angle of attack. The dynamics is given by

$$\dot{x} = \begin{bmatrix} -1.01887 & 0.90506 & -0.00215 \\ 0.82225 & -1.07741 & -0.17555 \\ 0 & 0 & -1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} d.$$

The ACD learning algorithm in Subsection V-B are determined by $(\alpha_a, \alpha_c, \alpha_d) = (0.3, 1.2, 0.1)$ with the bounds $(\psi_a, \psi_d) = (2, 3)$. The basis functions for parameterization are chosen as $\phi_c(x) = [x_1^2, x_1x_2, x_1x_3, x_2^2, x_2x_3, x_3^2]^\top$ and $\phi_a(x) = \phi_d(x) = x$. For comparison, we apply the conventional ETC design in [22] and a TTC baseline with a sampling period of 0.1s. Let the system evolve over $[0, 50]s$ with random initial state and initial weights.

As shown in Fig. 2, all state components are regulated toward the origin under the learned policies, while the sampled state $\hat{x}$ are updated only at the selected execution instants. Notably, the sampling becomes progressively sparser as the closed-loop system stabilizes. Fig. 3 further indicates that the achieved disturbance attenuation level under STC converges to a smaller steady-state value and stays below the prescribed bound $\gamma = 5$ (dashed line), whereas ETC and TTC converge to significantly larger levels, implying inferior disturbance rejection. In addition to the performance improvement, STC requires only 12 samples over the simulation horizon, compared with 24 for ETC and 500 for TTC, demonstrating substantial communication savings without sacrificing regulation quality.

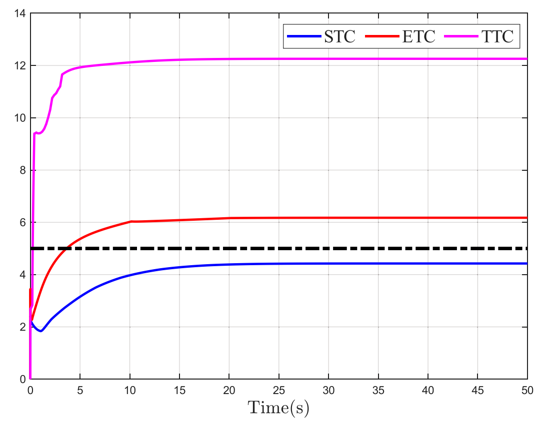

Fig. 3. Attenuation performance of STC, ETC and TTC.

B. Case 2: Nonlinear Single-Link Robot Arm.

The nonlinear single-link robot arm is widely adopted in practical industrial applications, with the dynamics [43]:

$$\ddot{\eta} + \frac{\mathbf{D}}{\mathbf{G}}\dot{\eta} + \frac{\mathbf{MgH}}{\mathbf{G}}\sin(\eta) - \frac{1}{\mathbf{G}}(u + d) = 0,$$

where η, u, and d are the joint angular, torque, and disturbance, respectively. Moreover, $\mathbf{D} = 2$ N is the viscous friction coefficient, $\mathbf{M} = 10$ kg is the payload mass, $\mathbf{g} = 9.81\,\mathrm{m/s^2}$ is the gravitational acceleration, $\mathbf{H} = 0.5$ m is the arm length, and $\mathbf{G} = 10\,\mathrm{kg \cdot m^2}$ is the moment of inertia. Define the state variables as $x_1 = \eta$, $x_2 = \dot{\eta}$, and $x = [x_1\ x_2]^\top$. Then, the state-space form of the robot arm is

$$\dot{x} = \begin{bmatrix} x_2 \\ -4.905\sin x_1 - 0.2x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0.1 \end{bmatrix} u + \begin{bmatrix} 0 \\ 1 \end{bmatrix} d.$$

The performance functional in (6) is specified by $Q(x) = \|x\|^2$, $R = 1$ and the attenuation level $\gamma = 10$. The players $\{\hat{u}_a, d_a\}$ in (27) are updated using the ACD learning algorithm presented in Subsection V-B with learning rates $(\alpha_a, \alpha_c, \alpha_d) = (0.5, 2, 0.2)$ and projection constraints $(\psi_a, \psi_d) = (2.5, 2)$. The basis functions for the ACD learning are chosen identically as $\phi_a(x) = \phi_c(x) = \phi_d(x) = [x_1^2, x_1x_2, x_2^2]^\top$. Moreover, the initial state and all adaptive weights are initialized by choosing each component uniformly from $(-1, 1)$.

To better demonstrate the effectiveness of the proposed self-triggered ACD learning scheme, we compare it with the ETC design in [22] and a conventional TTC baseline with a sampling

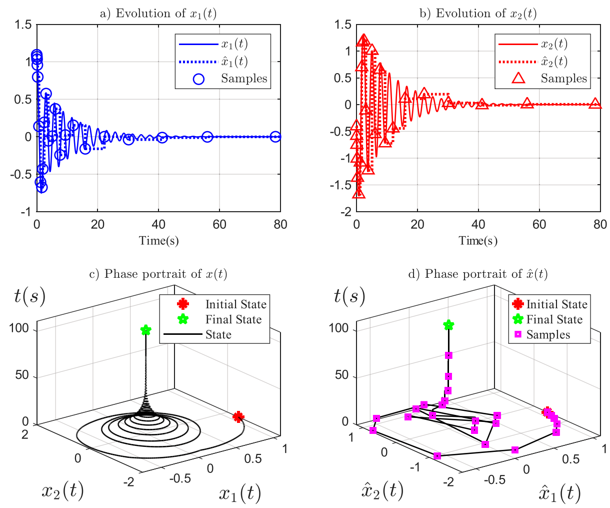

Fig. 4. Trajectories of the system state with 3D phase portraits.

TABLE III
PERFORMANCE METRICS UNDER DIFFERENT MECHANISMS

Metrics	STC	ETC	TTC
Samples	23	218	800
Average interval	4.8167s	0.3698s	0.1s
Minimal interval	0.05s	0.00002s	0.1s
Final $\|x(t)\|$	0.00003	0.00003	0.00004
Attenuation level	8.4019	12.9913	15.3096

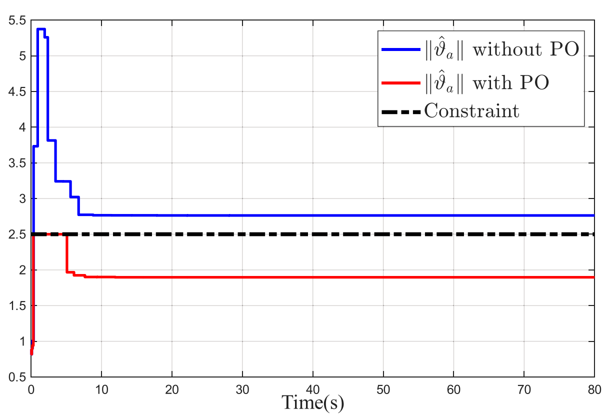

Fig. 5. Comparison of the adaptive weights with and without PO.

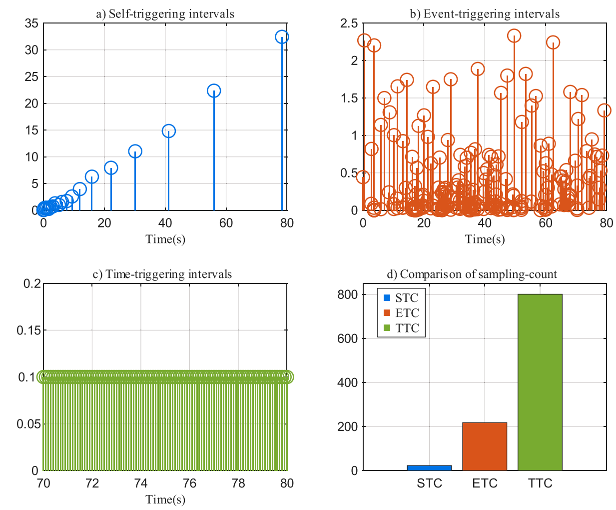

Fig. 6. Triggering intervals under STC, ETC, and TTC schemes, along with their comparison. For clarity, the TTC intervals are shown only over [70,80]s.

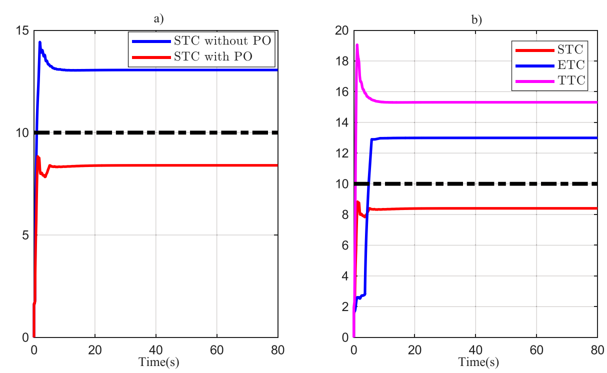

Fig. 7. Attenuation level. a) Comparison between with and without PO. b) Comparison among STC, ETC and TTC.

period of 0.1 s. Under these settings, the results of the proposed method are shown in Figs. 4 to 7 and Table III.

The trajectories of $x_1(t)$ and $x_2(t)$ in Fig. 4 decay to a small neighborhood of the origin, and the 3D phase portrait shows a clear convergence trend from the initial state to the final state, which is consistent with Theorem 3 that guarantees closed-loop stability under sampled-data implementation. The sparsity of samples as time increases matches the Zeno-free result in Theorem 1 implied by the self-triggered design (26).

Fig. 5 shows that, without PO, the Actor-weight norm exceeds the prescribed bound, whereas PO keeps it strictly within the constraint, which confirms Lemma 2 and highlights the necessity of bounded adaptive parameters for stability.

As depicted in Fig. 6, the self-triggered intervals grow as the system stabilizes (see Fig. 4), indicating that the CI-oriented rule computes progressively longer inter-execution times while maintaining regulation. The event-triggered intervals fluctuate and can become very small, reflecting the need for frequent monitoring and the practical risk of near-Zeno behavior. The comparison plot in Fig. 6(d) confirms the communication and computation advantages, i.e., STC uses far fewer samples than ETC and TTC for the same horizon.

In Fig. 7(a), under STC, adding PO reduces the achieved attenuation level below the prescribed γ threshold (dashed line), whereas the "without PO" case violates the H_∞ requirement (10). As drawn in Fig. 7(b), STC achieves the smallest steady attenuation level compared with ETC and TTC. This indicates that the proposed self-triggered learning achieves disturbance attenuation with minimal energy, demonstrating strong suitability for resource-constrained settings.

As summarized in Table III, the CI-based STC mechanism attains essentially the same regulation accuracy as ETC and TTC with fewer samples and a much larger average interval, which is advantageous for real-world deployment by reducing sensing/communication load and the frequency of online learning updates. The strictly positive minimal interval for STC is consistent with Theorem 1 ensuring Zeno-free execution, thereby avoiding the frequent triggering and scheduler overload that ETC may suffer when measurements are noisy or quantized. A practical challenge of STC is that each trigger requires slightly more on-board computation to predict the next sampling instant; however, this cost is paid only intermittently and is typically outweighed by the substantial reduction in total updates. The attenuation-level row is consistent with Fig. 7(b): STC satisfies the prescribed γ and achieves a smaller H_∞ level (10), suggesting better tolerance to real-world disturbances and implementation imperfections.

VII. Conclusion

An online projected ACD learning framework is developed to seek the Nash equilibrium of the zero-sum differential game, in which a CI-oriented self-triggered sampling rule is introduced. Exploiting the intelligence of self-triggering, the proposed scheme autonomously schedules update instants using sampled-state information, thereby eliminating continuous trigger monitoring and saving communication resources. Moreover, projection-based constraints confine the Actor and Disturbance weights within prescribed bounds, improving closed-loop reliability and disturbance attenuation performance. By jointly leveraging current and recorded data, the learning architecture reuses information more effectively, relaxes the conventional PE requirement, and enhances practical applicability. Future work will extend the proposed self-triggered ACD learning to fault-resilient control and model-free settings.

Appendix A
Proof of Theorem 2

First, a dissipative inequality about $\dot V^*$ is derived. With a slight abuse of notations, we write $u^*(x)$, $u^*(\hat x)$, and $d^*(x)$ simply as u^*, $\hat u^*$ and d^*, respectively. Then, noting that V^* is positive definite by the ZSO (9) and [36, Lemma 3.2.3], we consider it as the Lyapunov candidate for the system (5), whose time derivative under the input $(\hat u^*, d^*)$ is

$$\dot V^*(x) = \nabla^\top V^*(x) F(x, \hat u^*, d^*).$$

With (16) and (17), we have

$$\begin{aligned}\dot V^*(x) &= -Q(x) + \|u^*\|_R^2 - \gamma^2\|d^*\|^2 - 2u^{*\top} R\hat u^* + 2\gamma^2\|d^*\|^2\\ &= -Q(x) + \|u^* - \hat u^*\|_R^2 - \|\hat u^*\|_R^2 + \gamma^2\|d^*\|^2.\end{aligned}$$

Furthermore, d^* is bounded on a compact set, i.e., there exists positive constant $d_{\max}$ s.t. $\|d^*\| \le d_{\max}$. By (18) and the fact that $q_1\|x\|^2 \le Q(x)$, $\dot V^*(x)$ further satisfies the following dissipative inequality

$$\begin{aligned}\dot V^*(x) \le\ & -q_1\|x\|^2 + \mathcal{L}_u^2\bar\lambda(R)\|\tilde x\|^2 - \|\hat u^*\|_R^2 + \gamma^2\|d^*\|^2\\ =\ & -q_1\beta\|x\|^2 - \mu\|\hat u^*\|_R^2 + \gamma^2\delta\|d^*\|^2\\ & -q_1\bar\beta\|x\|^2 - \bar\mu\|\hat u^*\|_R^2 + \gamma^2\bar\delta d_{\max}^2 + \mathcal{L}_u^2\bar\lambda(R)\|\tilde x\|^2\end{aligned} \tag{53}$$

with constants $\beta, \mu, \delta \in (0,1)$ and their conjugates $\bar\beta = 1-\beta$, $\bar\mu = 1-\mu$, and $\bar\delta = 1-\delta$.

Next, we deal with the term $\|\tilde x\|$ in (53). Since $\kappa_1, \kappa_2 \in \mathcal{K}$, substituting (24) and (25) into (26) yields $\kappa_2(\|\tilde x\|^2) \le \mu_1\kappa_1(\|x\|^2 + \mu_2)$, and then by the convexity of κ_1,

$$\mu_1\kappa_1(\|x\|^2 + \mu_2) \le \frac{\mu_1}{2}\kappa_1(2\|x\|^2) + \frac{\mu_1}{2}\kappa_1(2\mu_2).$$

Since κ_1 and κ_2 are class $\mathcal{K}$ functions with the properties that $r \le \kappa_2(r)$ and $\kappa_1(r) \le r$, the above inequality further implies

$$\|\tilde x\|^2 \le \mu_1\|x\|^2 + \frac{\mu_1}{2}\kappa_1(2\mu_2). \tag{54}$$

Select the parameters in (54) s.t.

$$\mu_1 = \frac{q_1\bar\beta}{\mathcal{L}_u^2\bar\lambda(R)} > 0,\quad \frac{\mu_1}{2}\kappa_1(2\mu_2) = \frac{\mu_3}{\mathcal{L}_u^2\bar\lambda(R)} > 0 \tag{55}$$

where $\mu_3 \triangleq \bar\mu\,\|\hat u^*\|_R^2 - \gamma^2\,\bar\delta\, d_{\max}^2 > 0$ holds by $\delta \ge 1 - \frac{\bar\mu\underline{\lambda}(R)u_{\min}^2}{\gamma^2 d_{\max}^2}$ with $u_{\min}$ being the lower bound of $\|\hat u^*\|$ on the compact set s.t. $u_{\min} \le \|\hat u^*\|$.

Therefore, the fact in (54) implies that

$$\|\tilde x\|^2 \le \frac{q_1\bar\beta}{\mathcal{L}_u^2\bar\lambda(R)}\|x\|^2 + \frac{\mu_3}{\mathcal{L}_u^2\bar\lambda(R)}.$$

Then, $\dot V^*$ can be upper bounded as

$$\dot V^*(x) \le -q_1\beta\|x\|^2 - \mu\|\hat u^*\|_R^2 + \gamma^2\delta\,\|d^*\|^2. \tag{56}$$

Finally, the impact of d^* on the stability of the system (5) can be discussed in the following two cases.

1) If $d^* = 0$, then (56) becomes

$$\dot V^*(x) \le -q_1\beta\|x\|^2 - \mu\|\hat u^*\|_R^2 \le -q_1\beta\|x\|^2, \tag{57}$$

which indicates exponential stability of the system (5) at the origin. By Assumption 1 and [37, Theorem 5.1], we can obtain the finite-gain $\mathcal{L}_2$ stability of the system (5) from (57).

2) If $d^* \ne 0$ but $d^* \in \mathcal{L}_2$, for any $T > 0$ one has

$$\int_0^T \dot V^*(x(t))dt = V^*(x(T)) - V^*(x_0).$$

Hence, integrating both sides of (56) on $[0,\infty)$ yields

$$\int_0^\infty \|\hat y\|^2 dt \le \gamma^2\int_0^\infty \|d^*\|^2 dt + \frac{V^*(x_0)}{\delta}, \tag{58}$$

where $\hat y \triangleq \sqrt{(q_1\beta\|x\|^2 + \mu\|\hat u^*\|_R^2)/\delta}$.

Recall the performance output in (6). By selecting the parameters q_1, q_2, β, μ, δ s.t.

$$1 - \frac{\bar\mu\underline{\lambda}(R)u_{\min}^2}{\gamma^2 d_{\max}^2} \le \delta \le \min\left\{\frac{q_1\beta}{q_2}, \mu\right\},$$

one can ensure $\|y\|_{\mathcal{L}_2} \le \|\hat y\|_{\mathcal{L}_2}$. Finally, with the fact that $\sqrt{a^2+b^2} \le |a| + |b|$, and (58), one has

$$\|y\|_{\mathcal{L}_2} \le \|\hat y\|_{\mathcal{L}_2} \le \gamma\|d^*\|_{\mathcal{L}_2} + \sqrt{V^*(x_0)/\delta}, \tag{59}$$

which corresponds to the attenuation condition (2) and further implies the finite-gain $\mathcal{L}_2$ stability of the system (5).

Appendix B
Proof of Theorem 3

From (5), (36), (40), (42), and (46), the impulsive dynamics of the augmented state $\mathcal{X} = (x, \hat{x}, \tilde{\vartheta}_c, \tilde{\vartheta}_a, \tilde{\vartheta}_d)$ consists of

1) Continuous dynamics: over each time interval $[t_i, t_{i+1})$,

$$\dot{\mathcal{X}} = \Big(F(x, \hat{u}_a, d_a),\, 0,\, \alpha_c \nabla E_c,\, 0, -\alpha_d[(\phi_d\phi_d^\top + \mathfrak{D}_1)\tilde{\vartheta}_d - \mathfrak{D}_2]\Big).$$

2) Discrete dynamics: at each time $t = t_{i+1}$,

$$\mathcal{X}^+ = \mathcal{X}(t_i) + \Big(0,\, \tilde{x}(t),\, 0,\, \widehat{\vartheta}_a^*(t) - \hat{\theta}_a^+(t),\, 0\Big).$$

Here, we denote $[x^\top\, y^\top]^\top$ as (x, y). For stability analysis of this impulsive system, we construct the Lyapunov candidate

$$\mathcal{V}(\mathcal{X}) \triangleq V^*(x) + V^*(\hat{x}) + V_c(\tilde{\vartheta}_c) + V_a(\tilde{\vartheta}_a) + V_d(\tilde{\vartheta}_d).$$

(Stability of Continuous Dynamics) First, we focus on the time interval $[t_i, t_{i+1})$ between triggering instants, i.e., the continuous part. Taking the time derivative of $\mathcal{V}(\mathcal{X}(t))$ gives

$$\dot{\mathcal{V}}(\mathcal{X}) = \dot{V}^*(x) + \dot{V}^*(\hat{x}) + \dot{V}_c(\tilde{\vartheta}_c) + \dot{V}_a(\tilde{\vartheta}_a) + \dot{V}_d(\tilde{\vartheta}_d)$$

where $\dot{V}^*(\hat{x}) = \dot{V}_a(\tilde{\vartheta}_a) = 0$ since $\hat{x}(t)$ and $\tilde{\vartheta}_a(t)$ are updated only at $t = t_{i+1}$ but remain constant $\forall t \in [t_i, t_{i+1})$. That is,

$$\dot{\mathcal{V}}(\mathcal{X}) = \dot{V}^*(x) + \dot{V}_c(\tilde{\vartheta}_c) + \dot{V}_d(\tilde{\vartheta}_d). \tag{60}$$

1) Apply [37, Theorem 4.18] to $\dot{V}^*(x)$ in (35); then, $\exists\, \mu_x \in (0,1)$ and $T_x > 0$ s.t. for all $t \geq T_x$,

$$\|x(t)\| \leq \sqrt{\nu/(c_1\mu_x)} \triangleq x_{\max}$$

where ν denotes $\omega_0 c_2/(q_1\alpha)$. Recall (35). The ultimate bound $x_{\max}$ is mainly tuned by μ_x, α, and ω_0: increasing μ_x and α (both chosen close to 1) or decreasing ω_0 reduces $x_{\max}$. Specifically, ω_0 can be made smaller by lowering its upper bound ω through smaller R, ψ_a and ψ_d (see Subsection V-C1). Then, the tighter bound for $\|x(t)\|$ can be achieved.

By the above bound on x, the Critic residual ϵ_c in (38) is also bounded for all $t \geq T_x$ as

$$\|\epsilon_c(t)\| \leq \sum_{i=1}^{3} \epsilon_{i,\max} + \bar{\varepsilon}_c' b_f x_{\max} \triangleq \bar{\epsilon}_c.$$

2) Using the above bound of ϵ_c along with the fact that $\frac{r}{1+r^2} \leq \frac{1}{2}$, $\forall r \in \mathbb{R}$, it follows that $\mathfrak{C}_2$ defined by (40) is bounded as $\|\mathfrak{C}_2(t)\| \leq \bar{\epsilon}_c(1 + L_c)/2, \forall t \geq T_x$.

Since $\bar{\sigma}_c$, with its chosen sample points $\{\bar{\sigma}_{\mathfrak{c}_j}\}_j$, is assumed to satisfy the SE condition, there exists $\beta_c > 0$ s.t. $\underline{\lambda}(\mathfrak{C}_1) \geq \beta_c$, indicating that $\dot{V}_c(\tilde{\vartheta}_c)$ in (41) can be further bounded as

$$\dot{V}_c(\tilde{\vartheta}_c) \leq -\tilde{\vartheta}_c^\top \bar{\sigma}_c \bar{\sigma}_c^\top \tilde{\vartheta}_c - \beta_c\|\tilde{\vartheta}_c\|^2 + \|\tilde{\vartheta}_c\| \cdot \|\mathfrak{c}_2\|.$$

Therefore, by [37, Theorem 4.18], there exists a finite time $T_c > T_x > 0$ and a constant $\mu_c \in (0,1)$ s.t. for all $t \geq T_c$,

$$\|\tilde{\vartheta}_c(t)\| \leq \bar{\epsilon}_c(1 + L_c)/(2\beta_c\mu_c) \triangleq \tilde{\vartheta}_{c,\max}$$

and thus $\|\hat{\vartheta}_c(t)\| \leq \|\vartheta_c\| + \tilde{\vartheta}_{c,\max}$ $(\because \hat{\vartheta}_c(t) = \vartheta_c - \tilde{\vartheta}_c(t))$.

3) Substituting the above bound on $\hat{\vartheta}_c(t)$ and the projection bound $\|\hat{\vartheta}_d\| \leq \phi_d$ into the definition (32) of e_d and ϵ_d, one has

$$|e_d| \leq \psi_d\bar{\phi}_d + \frac{1}{2\gamma^2} b_k \bar{\phi}_c'(\|\vartheta_c\| + \tilde{\vartheta}_{c,\max}),$$

$$|\epsilon_d| \leq \|\vartheta_d\|\bar{\phi}_d + \frac{1}{2\gamma^2} b_k \bar{\phi}_c'(\|\vartheta_c\| + \tilde{\vartheta}_{c,\max})$$

after the time $T_c > 0$. From those bounds, $\mathfrak{D}_1$ defined by (47) can be also bounded after time T_c as $\|\mathfrak{D}_1\| \leq (1 + L_d)\xi_d$, where $\xi_d \triangleq (\|\vartheta_d\| + \psi_d)\bar{\phi}_d + \frac{1}{\gamma^2} b_k \bar{\phi}_c'(\|\vartheta_c\| + \tilde{\vartheta}_{c,\max})$.

Then, with ϕ_d being SE with its sample points $\{\phi_{\mathfrak{d}_j}\}_j$, there exists $\beta_d > 0$ s.t. $\underline{\lambda}(\mathfrak{D}_1) \geq \beta_d$, so that $\dot{V}(\tilde{\vartheta}_d)$ in (48) can be bounded after time T_c as

$$\dot{V}_d(\tilde{\vartheta}_d) \leq -\tilde{\vartheta}_d^\top \phi_d^\top \phi_d \tilde{\vartheta}_d - \beta_d\|\tilde{\vartheta}_d\|^2 + (1 + L_d)\xi_d \cdot \|\tilde{\vartheta}_d\|.$$

Therefore, by [37, Theorem 4.18], there exists a finite time $T_d > T_c > 0$ and a constant $\mu_d \in (0,1)$ s.t. for all $t \geq T_d$,

$$\|\tilde{\vartheta}_d(t)\| \leq (1 + L_d)\xi_d/(\beta_d\mu_d).$$

(Stability of Discrete Dynamics) We now turn our attention to the discrete dynamics at triggering instants $t = t_{i+1}$, where

$$\Delta V^*(x(t)) = \Delta V(\tilde{\vartheta}_c(t)) = \Delta V(\tilde{\vartheta}_d(t)) = 0.$$

Thus, the difference of $\mathcal{V}(\mathcal{X})$ can be expressed as

$$\Delta\mathcal{V}(\mathcal{X}) \triangleq \mathcal{V}(\mathcal{X}^+) - \mathcal{V}(\mathcal{X}) = \Delta V_a(\tilde{\vartheta}_a) + \Delta V^*(\hat{x}).$$

a) Consider only the instants $t = t_{i+1}$ after time $T_d > 0$ so that $\|\tilde{\vartheta}_c(t)\| \leq \tilde{\vartheta}_{c,\max}$, $\|x(t)\| \leq x_{\max}$, and $\|\tilde{\vartheta}_d(t)\| \leq \tilde{\vartheta}_{d,\max}$. Then, under Assumption 3, the term $\mathfrak{A}_2$ defined as (44) satisfies $\|\mathfrak{A}_2\| \leq L_a\xi_a$ with $\xi_a \triangleq \bar{\phi}_a^2\|\vartheta_a\| + \frac{1}{2}\bar{\phi}_a\underline{\lambda}(R)b_g\bar{\phi}_c'(\|\vartheta_c\| + \tilde{\vartheta}_{c,\max})$, which further contributes to the bounds of κ and ϖ as

$$\begin{aligned}
&|\kappa| \leq \kappa_{\max} \triangleq \\
&\bar{\omega}_1^2(\bar{\omega}_2^2/4 + \bar{\varepsilon}_a^2 + \bar{\omega}_2\bar{\varepsilon}_a) + \alpha_a L_a \xi_a \bar{\omega}_1\bar{\omega}_2 \\
&+ \alpha_a^2 L_a^2 \xi_a^2 + 2\alpha_a L_a \xi_a \bar{\varepsilon}_a \bar{\omega}_1, \\
&\|\varpi\| \leq \varpi_{\max} \triangleq \\
&\alpha_a(1 + L_a)\bar{\phi}_a^2(\bar{\omega}_1\bar{\omega}_2 + 2\bar{\omega}_1\bar{\varepsilon}_a + \alpha_a L_a \xi_a) \\
&+ \bar{\omega}_1\bar{\omega}_2 + 2\bar{\omega}_1\bar{\varepsilon}_a + 2\alpha_a L_a \xi_a,
\end{aligned}$$

where we have also used the bound $\|\mathfrak{A}_0 + \mathfrak{A}_1\| \leq (1 + L_a)\bar{\phi}_a^2$, and $\bar{\omega}_1$ and $\bar{\omega}_2$ are the upper bounds of ω_1 and ω_2 in (43): $\|\omega_1\| \leq \bar{\omega}_1 \triangleq \alpha_a\bar{\phi}_a$ and $|\omega_2| \leq \bar{\omega}_2 \triangleq \underline{\lambda}(R) \cdot (\bar{\phi}_c' \cdot \tilde{\vartheta}_{c,\max} + \bar{\varepsilon}_c')$.

Moreover, with $\hat{\phi}_a$ being SE with its sample points $\{\hat{\phi}_{\mathfrak{a}_j}\}_j$, there exists $\beta_a > 0$ s.t. $\underline{\lambda}(\mathfrak{A}_1) \geq \beta_a$. Therefore, the difference $\Delta V_a(\tilde{\vartheta}_a)$ given by (45) can be bounded as

$$\Delta V_a(\tilde{\vartheta}_a) \leq -(2\alpha_a\beta_a - \alpha_a^2\omega_3^2)\|\tilde{\vartheta}_a\|^2 + \varpi_{\max}\|\tilde{\vartheta}_a\| + \kappa_{\max} + \|p\|^2 - 2p^\top(\widehat{\vartheta}_a^* - \widehat{\vartheta}_a),$$

where $\omega_3 \triangleq \bar{\lambda}(\mathfrak{A}_0 + \mathfrak{A}_1)$; the p-term $\mathcal{P} \triangleq \|p\|^2 - 2p^\top(\widehat{\vartheta}_a^* - \widehat{\vartheta}_a)$ is always non-positive as shown below.

- If $\|\widehat{\vartheta}_a\| \leq \psi_a$, then $p = 0$ by definition, hence $\mathcal{P} = 0$.
- If $\|\widehat{\vartheta}_a\| > \psi_a$, then $p = \bar{\vartheta}_a - \widehat{\vartheta}_a$ for $\bar{\vartheta}_a \triangleq \psi_a\widehat{\vartheta}_a/\|\widehat{\vartheta}_a\|$ by its definition. Hence, substituting it into $\mathcal{P}$ yields

$$\mathcal{P} = -\|p\|^2 - 2(\bar{\vartheta}_a - \widehat{\vartheta}_a)^\top(\widehat{\vartheta}_a^* - \bar{\vartheta}_a) \leq -\|p\|^2 \leq 0,$$

where the first inequality comes by Lemma 2.

Based on the fact $\mathcal{P} \leq 0$ and the Young's inequality with $\upsilon_a > 0$: $\|\tilde{\vartheta}_a\|\varpi_{\max} \leq \frac{1}{2}\upsilon_a\|\tilde{\vartheta}_a\|^2 + \frac{1}{2}\upsilon_a^{-1}\varpi_{\max}^2$, $\Delta V_a(\tilde{\vartheta}_a)$ can be further upper-bounded as

$$\Delta V_a(\tilde{\vartheta}_a) \leq -\mathfrak{q}(\alpha_a)\|\tilde{\vartheta}_a\|^2 + \frac{\varpi_{\max}^2}{2\upsilon_a} + \kappa_{\max},$$

where $\mathfrak{q}(z) \triangleq 2\beta_a z - \omega_3^2 z^2 - \upsilon_a/2 \quad (z > 0)$ is quadratic in z.

Let $z = \alpha_a \in (0, 2\beta_a/\omega_3^2)$ and choose $\upsilon_a < 2\beta_a^2/\omega_3^2$ such that $z \in (z^-, z^+)$ for $z^\pm = (\beta_a \pm \sqrt{\beta_a^2 - \upsilon_a\omega_3^2/2})/\omega_3^2$ and thus $(z^-, z^+) \subset (0, 2\beta_a/\omega_3^2)$. This ensures that $\mathfrak{q}(\alpha_a) > 0$, and thus, by [44, Theorem 2.4.6], there exist constants $N_a \in \mathbb{N}_0$ and $\mu_a \in (0,1)$ s.t.

$$\|\tilde{\vartheta}_a(t_i)\| \leq \sqrt{\frac{\varpi_{\max}^2 + 2\upsilon_a\kappa_{\max}}{2\mu_a\upsilon_a \cdot \mathfrak{q}(\alpha_a)}} \quad \text{for all } i \geq N_a.$$

b) For the difference $\Delta V^*(\hat{x})$ given by (37), the application of [44, Theorem 2.4.6] ensures that there exists constants $N_{\hat{x}} \in \mathbb{N}_0$ and $\mu_{\hat{x}} \in (0,1)$ and s.t. for all $i \geq N_{\hat{x}}$,

$$\|\hat{x}(t_i)\| \leq \sqrt{c_2\eta_b/(c_1^2\mu_{\hat{x}}\eta_a)}.$$

(Stability of Hybrid Dynamics) Collecting all the preceding discussions and results, we conclude that the augmented hybrid state $\mathcal{X}$ is UUB, e.g., by [45, Corollary 2.5].

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